

TABLE 30.1 Particles of the Standard Model in the Relativistic (High-Temperature) Limit. See Eq. (29.72) and the accompanying discussion for n_{anti} and n_{spin} ; f is the fermion factor in Eq. (30.2).

Particle	# Types	n_{anti}	n_{spin}	g	f	Contribution to g_*
Photon	1	1	2	2	1	2.00
Charged leptons	3	2	2	12	7/8	10.50
Neutrinos	3	2	1	6	7/8	5.25
Quarks	18	2	2	72	7/8	63.00
Gluons	8	1	2	16	1	16.00
Vector gauge bosons	3	1	3	9	1	9.00
Scalar Higgs boson	1	1	1	1	1	1.00

TABLE 30.2 Eras and Events in the Early Universe. The values are approximate.

Era or Event	Time	Temperature (kT)
Planck era	$< 5 \times 10^{-44}$ s	$> 10^{19}$ GeV
Planck transition	5×10^{-44} s	10^{19} GeV
Grand unification era	5×10^{-44} s to 10^{-36} s	10^{19} GeV to 10^{15} GeV
Inflation	10^{-36} s to 10^{-34} s	10^{15} GeV
Electroweak era	10^{-34} s to 10^{-11} s	10^{15} GeV to 100 GeV
Electroweak transition	10^{-11} s	100 GeV
Quark era	10^{-11} s to 10^{-5} s	100 GeV to 200 MeV
Quark–hadron transition	10^{-5} s	200 MeV
Neutrino decoupling	0.1 s	3 MeV
Electron–positron annihilation	1.3 s	1 MeV

transition at about 10^{-5} s, when a plasma of free quarks and gluons condensed to form hadrons, including the familiar proton and neutron.

Table 30.2 shows the typical times and temperatures (kT) of different eras during the first second after the Big Bang. (We remind you that any “age of the universe” that is quoted has been obtained from a backward extrapolation in time, using a Friedmann model of the universe.) The temperature as a function of time is given by Eq. (29.89), which we reproduce here:

$$T(t) = (1.81 \times 10^{10} \text{ K s}^{1/2}) g_*^{-1/4} t^{-1/2}. \quad (30.1)$$

The value of g_* , the effective number of degrees of freedom of all of the relativistic particles present at temperature T , is given by

$$g_* = \sum_{\text{bosons } i} g_i + \frac{7}{8} \sum_{\text{fermions } i} g_i \left(\frac{T_i}{T} \right)^4. \quad (30.2)$$

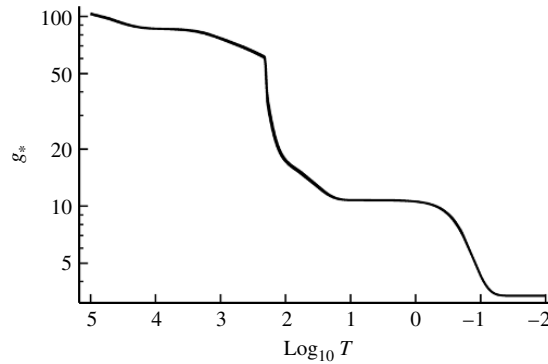


FIGURE 30.1 The effective number of degrees of freedom, g_* , as a function of temperature. The temperature is actually kT in units of MeV. The quark-hadron transition is chosen to occur at 200 MeV. (Figure adapted from Coleman and Roos, *Phys. Rev. D*, 68, 027702, 2003.)

The factor $(T_\nu/T)^4 = (4/11)^{4/3}$ (Eq. 29.73) is applied only to neutrinos, and only if the temperature is below that of neutrino decoupling ($kT \approx 3$ MeV).

Figure 30.1 shows the variation in the value of g_* as a function of kT in units of MeV. The present value of $g_* = 3.363$ (Eq. 29.76) at the right side of the graph is due to the photon and the three species of neutrinos and their antineutrinos. Between neutrino decoupling at about 3 MeV and ~ 100 MeV, the electron and positron are the only additional relativistic particles, yielding $g_* = 10.75$. Above about 200 MeV, the hadrons are dissociated into their component quarks and gluons, which contribute more to the value of g_* . Finally, by $kT \sim 300$ GeV (beyond the left side of the graph), all of the particles of the standard model are relativistic and $g_* = 106.75$. No new particles masses are predicted within the range of energies of the electroweak era, so we may adopt a value of $g_* \simeq 100$ throughout this time interval. (The electroweak era is sometimes referred to as the *great desert* because of the paucity of new particles.) At even earlier times, it is estimated for the simplest model that $g_* \simeq 160$. Fortunately, $T(t)$ is sensitive only to the fourth root of g_* , so precise values of g_* are not required for the purpose of estimation.

Hot and Cold Dark Matter

Missing from Table 30.2, of course, are the particles that make up dark matter. Determining the composition of dark matter is one of the greatest challenges facing cosmologists today. Some dark matter may consist of ordinary baryonic matter, although the total amount of baryonic matter (both shining and dark) cannot make up more than 4% of the total mass of the universe. The WMAP results and constraints imposed by the Big Bang nucleosynthesis of the light elements are inconsistent with much larger amounts of baryonic matter (see Fig. 29.14). The gravitational microlensing of stars in the Large Magellanic Cloud by MACHOs (massive compact halo objects) described in Section 24.2 may demonstrate that some dark baryonic matter is hiding in galactic halos, perhaps in the form of brown dwarfs or stellar-mass black holes. However, a statistical analysis of the microlensing events indicates that only about 19% of the mass of the Milky Way's dark matter halo can be explained by MACHOs.

Dark matter candidates are usually divided into two categories, **hot dark matter** (HDM) and **cold dark matter** (CDM). Hot dark matter consists of particles moving with relativistic velocities.¹ The leading candidates for hot dark matter are massive neutrinos (which are leptons). Recall from Section 10.3 that the best estimate for the upper limit to the electron neutrino's mass is $2.2 \text{ eV}/c^2$. Cold dark matter candidates are hypothetical particles that move slowly, such as the WIMPs (weakly interacting massive particles) that hardly interact with normal matter except through their gravitational attraction.² It has been suggested that the rest energy of WIMPs may range from 10 GeV (about 10 times more massive than a proton) up to several TeV (10^{12} eV). Another hypothetical cold dark matter candidate is the **axion**, a low-mass boson that is extremely lightweight ($mc^2 \approx 10^{-5} \text{ eV}$). If axions do exist, and if they constitute most of the dark matter, then they are by far the most numerous type of particle in existence. Although there is as yet no evidence for the existence of WIMPs or axions, the search for them continues.

The distinction between hot and cold dark matter is important because it is difficult for the relativistic hot dark matter to clump together gravitationally and participate in the formation of structure in the early universe. For this reason, as we shall see in Section 30.2, models that incorporate cold dark matter are currently favored. In fact, the standard model in cosmology is referred to as the **Λ CDM model** because it includes both the cosmological constant and cold dark matter.

The Planck Limits on Time, Mass, and Length

The earliest time that can be addressed by current physical theory is the **Planck time**,

$$t_P \equiv \sqrt{\frac{\hbar G}{c^5}} = 5.39 \times 10^{-44} \text{ s}. \quad (30.3)$$

The Planck time is the only combination of fundamental constants that has units of time and, as such, is a characteristic quantity in fundamental theories. It contains Planck's constant, Newton's gravitational constant, and the speed of light, a melding of quantum mechanics, gravitation, and relativity that has yet to be achieved in a unified theory.

The physicist's toolbox of theories cannot penetrate this barrier in time. Some insight into the limits imposed by the Planck time comes from Heisenberg's uncertainty principle, Eq. (5.19). Consider a black hole of mass M , the most compact region within which a mass can be contained. The central singularity is confined within the event horizon, so the uncertainty in its position is just its Schwarzschild radius,

$$\Delta x \approx R_S.$$

Thus the uncertainty in its momentum is

$$\Delta p \approx \frac{\hbar}{\Delta x} \approx \frac{\hbar}{R_S}.$$

¹More precisely, it is the internal velocity dispersion of the particles that is relativistic.

²WIMPs were previously discussed as a candidate for a component of the Milky Way's dark matter halo in Section 24.2.

For a very low-mass primordial black hole that formed immediately after the Big Bang, R_S will be extremely small and so Δp will be quite large. In the relativistic limit, the energy of the singularity is, from Eq. (4.48), $E = pc$, and so the uncertainty in the energy is

$$\Delta E = (\Delta p)c \approx \frac{\hbar c}{R_S}.$$

The expected value of the energy will be at least as large as ΔE , so the kinetic energy of the singularity is approximately $K \approx \hbar c/R_S$. There will be a conflict between the quantum and classical descriptions of gravity when this energy is comparable to the magnitude of the (Newtonian) gravitational energy of the black hole, which, according to Eq. (10.22), is approximately $U \approx -GM^2/R_S$. Is the black hole bound, or could the singularity emerge from behind the event horizon in violation of the “Law of Cosmic Censorship” discussed in Section 17.3? This confrontation between the application of quantum and classical physics to gravitation may be estimated to occur when the sum of the kinetic energy, K , and the potential energy, U , is zero:

$$\frac{\hbar c}{R_S} - \frac{GM^2}{R_S} = 0.$$

The resulting mass is called the **Planck mass**,

$$m_P \equiv \sqrt{\frac{\hbar c}{G}} = 2.18 \times 10^{-8} \text{ kg}. \quad (30.4)$$

As was mentioned in Problem 17.21, this is an estimate of the least massive primordial black hole. Inserting the Planck mass into Eq. (17.27) for the Schwarzschild radius, $R_S = 2GM/c^2$, neglecting the factor of two in this order-of-magnitude estimate, and solving for R_S results in a characteristic length called the **Planck length**,

$$\ell_P \equiv \sqrt{\frac{\hbar G}{c^3}} = 1.62 \times 10^{-35} \text{ m}. \quad (30.5)$$

The Planck time, $t_P = \ell_P/c$, is just the time required for light to travel the Planck length. The Planck length is therefore the diameter of a causally connected region (the horizon distance) at that time; see Eq. (29.154). Although the foregoing argument is only a little better than a dimensional analysis, it does reveal the significance of combining these fundamental constants to produce a characteristic time, mass, and length.

Equation (30.1) indicates that the temperature at the Planck time was nearly 10^{32} K, so the typical thermal energy of a particle was $kT \sim 10^{19}$ GeV! The description of the universe before the Planck time staggers even the strongest imagination. The universe would have been a collection of primordial black holes that were continually forming, evaporating, and reforming. In the process, different regions of spacetime were rapidly connecting and disconnecting, giving it a foamlike structure. Our current physical theories break down at times earlier than t_P , and in fact the very notion of space and time as separate concepts dissolves before the Planck time. A quantum theory of gravity capable of describing this

convoluted arena in which space and time have lost their familiar, separate identities has yet to be invented. After the Planck time, spacetime began to take on a more coherent structure as greater portions of it became causally connected. Exactly how time itself emerged from the Big Bang is a question to be pondered by physicists and philosophers alike.³

Unification and Spontaneous Symmetry Breaking

It is an article of faith for physicists that before the Planck time, the four fundamental forces of nature (the gravitational force, the electromagnetic force, and the strong and weak nuclear forces) were merged into one all-encompassing **Theory of Everything** (TOE); see Fig. 30.2. Although the rough outlines of such a theory are still a matter of conjecture, the theory must have certain mathematical symmetries guaranteeing that the four forces were conjoined. When the universe reached the Planck time, the single, all-encompassing TOE force spontaneously separated into the gravitational force (as described by Einstein's theory of general relativity) and a unified version of the three remaining forces. This process is called **spontaneous symmetry breaking**, a term that refers to the changes in the mathematical symmetries of the theory's equations.⁴

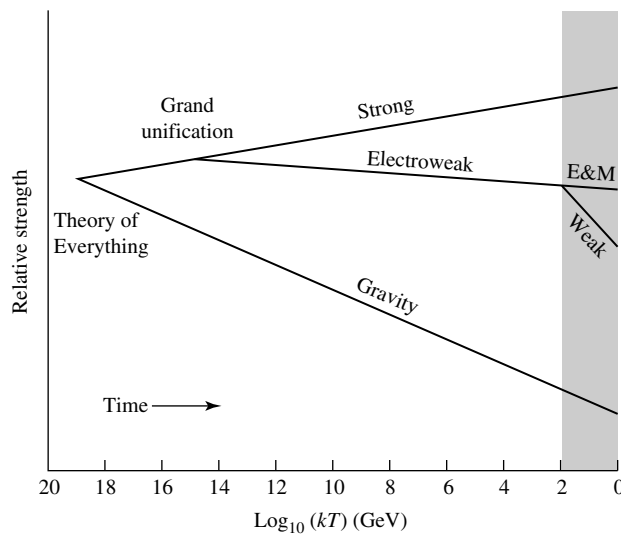


FIGURE 30.2 The unification of the four fundamental forces, according to a Theory of Everything. The electromagnetic and weak forces were unified at times earlier than 10^{-11} s, and the electroweak and strong forces were unified before 10^{-34} s. All four forces were joined at times earlier than the Planck time, $t_p = 5 \times 10^{-44}$ s. “E&M” stands for the electromagnetic force. Only the electroweak unification has been tested experimentally (shaded region).

³For example, is the statement that “time was created by the Big Bang” meaningful? Because any act of creation involves a time sequence (“first it isn’t, then it is”), can time itself be created?

⁴A physical example of spontaneous symmetry breaking is a marble rolling around the rim of a roulette wheel. Initially, the marble is equally likely to be found above any of the numbered bins, but as the marble loses energy it falls into one of them and becomes trapped.

The mathematics involved in spontaneous symmetry breaking is similar to that which describes a familiar phase transition in ordinary matter. For example, above a temperature of 273 K, water is in a liquid state. Assuming that there is no preferred direction imposed from outside, the water molecules are randomly oriented in all directions. When the temperature falls below 273 K, however, the water undergoes a phase transition and freezes. The symmetry of the liquid water is broken because the arrangement of H_2O molecules in the lattice of an ice crystal has preferred directions along the axes of the lattice. Just as there is a release of latent heat when water freezes, energy was released in the episodes of spontaneous symmetry breaking in the early universe.

The theories (there are several variants) that describe the joining of the remaining three forces are called **grand unified theories** (GUTs). The simplest GUT, proposed in 1973 by the American physicists Sheldon Glashow and Howard Georgi (both at Harvard), is known as $\text{SU}(5)$. The GUTs have had some successes, such as providing a fundamental explanation for the equal magnitudes of the proton and electron charge. They have also had some setbacks; the failure to detect $\text{SU}(5)$'s predicted decay of the proton probably eliminates it as a successful GUT. The moral is that specific predictions based on grand unified theories should be viewed with some caution.

The unification of the strong and weak nuclear forces and the electromagnetic force would have lasted until the temperature of the universe had fallen to about 10^{29} K, when the characteristic thermal energy (kT) of a particle was about 10^{15} GeV and the universe was some 10^{-36} s old. At this point, following another episode of spontaneous symmetry breaking, the strong nuclear force parted company with the electroweak (combined electromagnetic and weak) force.

The theory of the electroweak unification was worked out in the 1960s by three physicists: Sheldon Glashow and Steven Weinberg (American) and Abdus Salam (Pakistani). They described how electromagnetic and weak forces were united when the temperature exceeded about 2×10^{15} K, at roughly 10^{-11} s.⁵ At this temperature, the characteristic thermal energy of a particle is about 150 GeV. Their theory predicted the existence of three new particles (the three vector gauge bosons $W^\pm$ and Z^0) that mediate the weak force, just as photons convey the electromagnetic force. Above $kT \sim$ a few hundred GeV, the vector gauge bosons become massless, and so become indistinguishable from photons. The electromagnetic and weak forces are then unified. When the temperature falls below about 2×10^{15} K, a spontaneous symmetry breaking endows the vector gauge bosons with mass. When these particles were discovered in the 1980s, the agreement between experiment and theory provided a striking confirmation of the electroweak unification. This success provides encouragement to physicists working on GUTs and TOEs.

Problems with the Standard Theory of the Big Bang

We now return to the Planck time to discover how spontaneous symmetry breaking, with its concomitant release of specific heat, affected the expansion of the early universe. As we move forward from the Planck time, we are confronted by three problems with the simple picture of the Big Bang we have considered so far.

⁵This is the earliest time that can be probed experimentally.

- *Why is the cosmic background radiation so smooth?* Consider two CMB photons arriving at the observer from opposite sides of the sky. It would seem that the regions from which photons came could never have communicated with each other, since the photons have just passed each other. In fact, the problem is even more severe. At the time of decoupling, regions of the CMB that are now separated by more than 2° on the sky could never have been in causal contact (see Problem 30.11). The spacetime diagram shown in Fig. 30.3 illustrates the situation. Photons arriving now at Earth, Point *A*, were emitted at the time of recombination from sources at Points *B* and *C*, which lie on the past light cone of Point *A*. However, when extended back to the Big Bang, the past light cones of Points *B* and *C* do not intersect. Points *B* and *C* are not in causal contact, since each is outside the other's particle horizon. The smoothness of the CMB is then very difficult to explain, especially since the variations that have been found are so very small ($\delta T/T \sim 10^{-5}$). The problem of finding the mechanism that allowed the universe's left hand to know what its right hand was doing is called the **horizon problem**.
- *Why is the universe so nearly flat* ($\Omega_0 \simeq 1$)? Recall from Example 29.1.1 that if Ω_0 is roughly of order unity, then at very early times ($z \gg 1$) the density parameter was essentially equal to one. For the simple model of a universe of pressureless dust considered in Section 29.1,

$$\frac{1}{\Omega_0} - 1 = \left(\frac{1}{\Omega} - 1 \right) (1 + z) \quad (30.6)$$

(see Problem 29.10). If Ω had differed slightly from one in the very early universe, today's value, Ω_0 , would vary significantly from unity. Suppose, at the time of decoupling ($[t_{\text{dec}}]_{\text{WMAP}} = 379 \text{ kyr}$ at $[z]_{\text{WMAP}} = 1089$), that the density parameter had

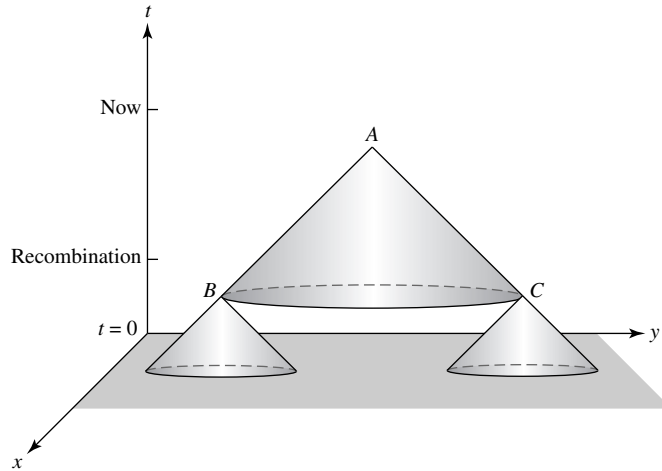


FIGURE 30.3 The horizon problem. The same CMB spectrum is observed at Point *A* from Points *B* and *C*, although *B* and *C* have never been in causal contact.

differed from unity by just 0.0009. If $\Omega_{\text{dec}} = 0.9991$, then according to Eq. (30.6), today's value of the density parameter would be $\Omega_0 \simeq 0.5$; and if $\Omega_{\text{dec}} = 1.0009$, then $\Omega_0 \simeq 50$. Such underdense and overdense universes have profound dynamical implications. For example, if $\Omega_{\text{dec}} = 0.5$, then after another few hundred thousand years the universe and its contents would be too widely dispersed and expanding too fast to allow the formation of galaxies or stars. On the other hand, if $\Omega_{\text{dec}} = 2$, then after a few hundred thousand years the expansion would have reversed itself, and the universe would have ended in a Big Crunch. Our existence on Earth reflects this fine-tuning of $\Omega = 1$ in the early universe. Otherwise, we would not be here today to wonder about it! The problem of the origin of our nearly flat universe is called the **flatness problem**.

- *Why have we found no magnetic monopoles?* A magnetic monopole is a single magnetic charge, tantamount to an isolated magnetic pole. When the universe experienced an episode of spontaneous symmetry breaking at the end of the GUTs era of unification, a choice of “direction” was made at each point in space, associated with the quantum field involved (called the Higgs field). This direction was not one of spatial orientation but, rather, involved a choice of certain discrete values (analogous to quantum numbers) for the theory describing the field. **Defects** arose at locations where a discontinuity in this choice of values existed, and a pointlike discontinuity corresponds to a **magnetic monopole**.⁶

Although a magnetic monopole might be $\sim 10^{16}$ times more massive than a proton ($mc^2 \sim 10^{16}$ GeV), there was sufficient energy at the end of the GUTs era to create copious numbers of magnetic monopoles. Yet despite many searches for magnetic monopoles, only one experiment has found any evidence for them, and then only one possible monopole candidate was detected (on Valentine's Day in 1982). There have not been any other detections to confirm that result. If our understanding of grand unification is correct in predicting the creation of magnetic monopoles, then where are they? This is called the **monopole problem**.

By itself, the Big Bang explanation for the beginning of the universe is incapable of meaningfully answering these questions. Of course, one possible answer is “That's just the way it is”—that the universe started out smooth and flat because of some set of initial conditions at $t = 0$. Cosmologists would prefer to find a natural sequence of events that inevitably led to the universe as it appears today, instead of specifying, in an ad hoc way, the values of certain variables at the beginning to make everything come out right.

Inflation

In 1980 an American astronomer working at Stanford University, Alan Guth, proposed a single solution. According to Guth, the Big Bang picture is essentially correct, but during its first fraction of a second, when t was barely greater than zero, the whole universe was much more compact than is described by the standard Big Bang. At that time, every point was close

⁶Defects can arise any time symmetry breaking occurs, but those from the end of the GUTs episode have received the most attention.

enough to every other point to be in causal contact, and the entire universe had achieved thermodynamic equilibrium. Then there was a tremendous spurt of *exponential expansion* that smoothed out the universe, rendering it exactly flat with $\Omega_0 = 1$ (or *very* nearly so). Guth called this period of exponential growth **inflation**. This episode of inflation would explain the general isotropy of the universe, and the smoothness of the cosmic background radiation in particular. After inflation, the expansion of the universe proceeded as in the standard Big Bang model.

The details of Guth's inflationary proposal involve ideas at the frontiers of particle physics. In fact, many variants on the original inflationary scenario have been proposed. As observations accumulate, one model may fall out of favor, perhaps to be revived later as more information becomes available. Much research is still ongoing in this area, and the version of inflation described in the following discussion may well be wrong in some of its specifics. However, most cosmologists believe that, because it gives such satisfying answers to so many questions, some form of inflation must have taken place in the early universe.

Virtual Particles and the Energy of the Vacuum

The particle energies of $\sim 10^{19}$ GeV at the Planck time were so great that the familiar baryons (such as protons, neutrons, and their antiparticles) present at much lower temperatures were absent. We need not concern ourselves with the highly relativistic particles that were present then. Instead, we will concentrate on another significant component of the very early universe, the energy of the **vacuum**. The term *vacuum* requires some explanation. In everyday language, it means a complete emptiness, devoid of any matter or energy. However, physicists use *vacuum* to describe the ground state of a system. For instance, in Example 5.4.2 an electron confined to a small region was calculated to have a certain minimum energy, its ground-state energy. This energy cannot be lost or extracted from the electron, since there is no lower quantum state to which the electron can make a transition. The existence of the vacuum has been verified by observations of the **Casimir effect** (named for Dutch physicist Hendrick Casimir, 1909–2000). Two uncharged parallel flat conducting plates with a very small separation will alter the properties of the vacuum between the plates. This change in the vacuum creates an attractive force between the plates, which has been measured. Unfortunately, the Casimir effect cannot be used to calculate the value of the energy density of the vacuum.

We can crudely estimate the value of the energy density of the vacuum using the uncertainty principles $\Delta x \Delta p \approx \hbar$ (Eq. 5.19) and $\Delta E \Delta t \approx \hbar$ (Eq. 5.20). The vacuum can be modeled as a place where matter–antimatter pairs of particles are constantly being created and annihilated. These particles cannot be directly observed and their energy cannot be tapped; they are known as **virtual particles**. They borrow their rest energy ΔE from the vacuum and are annihilated in such a short time Δt that they escape detection.

Let's consider a virtual particle of mass $m \approx (\Delta E)/c^2$ confined to a cubical box of side $L \approx \Delta x$ with a particle lifetime of

$$\Delta t \approx \hbar/\Delta E \approx \hbar/mc^2.$$

In addition, the particle's speed is approximately

$$v \approx \frac{\hbar}{m\Delta x} \approx \frac{\hbar}{mL}$$

(see Example 5.4.2). Since the farthest the particle can travel in time Δt is $v\Delta t$, we set $L = v\Delta t$ to be certain the particle's motion does not carry it outside the box. Thus

$$L = v\Delta t \approx \frac{\hbar}{mL} \frac{\hbar}{mc^2}.$$

Solving for L , we find

$$L \approx \frac{\hbar}{mc}.$$

The energy density of the vacuum must be capable of creating a pair of particles in the box so that conservation rules can be maintained (such as electric charge). That is, the energy density of the vacuum must be at least

$$u_{\text{vac}} \approx \frac{2mc^2}{L^3} \approx \frac{2m^4c^5}{\hbar^3}.$$

The greatest mass for each particle in the pair may be taken to be the Planck mass, $m_P = \sqrt{\hbar c/G}$ (Eq. 30.4). Our estimate of the energy density of the vacuum is thus

$$u_{\text{vac}} \approx \frac{2m_P^4c^5}{\hbar^3} \approx \frac{2c^7}{\hbar G^2},$$

or $u_{\text{vac}} \approx 10^{114} \text{ J m}^{-3}$. Of course this is only a rough estimate. More sophisticated calculations (which also invoke an arbitrary cut-off to avoid an infinite answer) result in a value of $u_{\text{vac}} \approx 10^{111} \text{ J m}^{-3}$.

Einstein's general relativity is responsive to every form of energy, including the energy of the vacuum. Furthermore, the energy density of the vacuum is a *constant*, just as for dark energy. The fluid equation (Eq. 29.50) shows that the constant vacuum energy density would produce a negative pressure

$$P_{\text{vac}} = -\rho_{\text{vac}}c^2 = -u_{\text{vac}}, \quad (30.7)$$

an equation of state with $w = -1$ (cf. Eq. 29.52). It is natural to identify dark energy as the energy density of the vacuum. However, the value of the vacuum energy density is huge compared to the energy density of dark energy. From Eq. (29.113),

$$u_{\text{dark}} \equiv \rho_{\Lambda}c^2 = \rho_{c,0}\Omega_{\Lambda,0}c^2 = 6.22 \times 10^{-10} \text{ J m}^{-3}. \quad (30.8)$$

If we identify dark energy as the energy density of the vacuum, we are faced with a daunting discrepancy of some 120 orders of magnitude between the theoretical and observed values. The huge value calculated for the vacuum energy density is in sharp conflict with the observed rate of the Hubble flow. If every cubic meter were accompanied by approximately

10^{111} J of vacuum energy as the universe expanded and its volume grew, the universe would have expanded so rapidly that no galaxies or stars could ever have coalesced under the influence of gravity.

A plausible physical mechanism for reducing the value calculated for today's vacuum energy density to the observed value of u_{dark} has yet to be found. According to some advanced particle theories, bosons and fermions should make contributions to the vacuum energy of opposite signs, and so cancel to yield zero vacuum energy. If the cancelation were not perfect, a small observed residual vacuum energy density could result. Why should the cancelation be effective for the first 120 decimal places and then break down? So far the combined efforts of cosmologists and particle physicists have shed little light on this mystery. Nevertheless, we will continue to identify dark energy as the energy density of the vacuum and to accept its presently observed value of nearly zero.

The False Vacuum

At the end of the GUTs epoch, when $t \sim 10^{-36}$ s and the temperature was $T \sim 10^{28}$ K, the universe entered an extremely peculiar state called the **false vacuum**. The false vacuum that existed when the universe was approximately 10^{-36} s old was not a true vacuum, so the universe was not in the state with the lowest possible energy density. Instead, the universe had entered a supercooled state in which its temperature had plummeted *far below* 10^{28} K, the temperature for spontaneous symmetry breaking. Supercooling happens when a phase transition proceeds much more slowly than the cooling rate. For example, water can be supercooled to more than 20 K below its freezing point without changing phase. Since freezing would release the water's latent heat, there is a state of lower energy density (ice) available to the supercooled water. However, the supercooled water persists in a liquid state of higher energy density. Similarly, the universe persisted in its false vacuum state of unbroken symmetry with a high energy density, even though a spontaneous symmetry breaking to a true vacuum with zero energy density was energetically favorable. According to grand unified theories, the false vacuum had a constant energy density on the order of⁷

$$u_{\text{fv}} \approx 10^{105} \text{ TeV m}^{-3} = 1.6 \times 10^{98} \text{ J m}^{-3}. \quad (30.9)$$

Quantum Fluctuations and the Onset of Inflation

When $t < 10^{-36}$ s, the universal expansion was governed by relativistic particles. Inflation began when quantum fluctuations governed by the uncertainty principle, Eq. (5.20), allowed a small region of space to enter a true vacuum state in a universe otherwise filled with false vacuum. Although the pressure within the bubble of true vacuum was essentially zero, it was surrounded outside by the negative pressure of the false vacuum. The greater pressure inside the bubble caused the bubble to grow at an astounding rate. Thus at roughly $t_i = 10^{-36}$ s, the constant energy density of the false vacuum became dominant in the acceleration equation,

⁷It is important to recall that dark energy plays an insignificant role at the time of inflation. If dark energy is identified as the energy density of the vacuum, then the mechanism of inflation requires an additional source of vacuum energy, called the **inflaton field**, that is depleted (or nearly so) by the time inflation ends.

Eq. (29.116). This equation then became of the form

$$\frac{d^2 R}{dt^2} = \left[-\frac{4}{3}\pi G \left(\frac{u_{fv}}{c^2} + \frac{3P_{fv}}{c^2} \right) \right] R. \quad (30.10)$$

Using $P_{fv} = -u_{fv}$ from Eq. (30.7), this becomes

$$\frac{d^2 R}{dt^2} = \frac{8\pi G u_{fv}}{3c^2} R. \quad (30.11)$$

The exponentially growing solution is

$$R(t) = R(t_i) e^{t/\tau_i}, \quad (30.12)$$

where $R(t_i)$ is the initial value of the scale factor when inflation began, and τ_i is the *inflation time scale* given by

$$\tau_i = \sqrt{\frac{3c^2}{8\pi G u_{fv}}} \approx 10^{-36} \text{ s}. \quad (30.13)$$

Over what time interval Δt did inflation occur? Fundamental physics cannot yet supply the answer, but we can estimate the *minimum* time interval by examining what inflation accomplished: allowing our present universe to be causally connected with a uniform cosmic microwave background. At the initial time $t_i = 10^{-36}$ s, the horizon distance was only

$$d_i = 2ct_i = 6 \times 10^{-28} \text{ m}, \quad (30.14)$$

from Eq. (29.154). By the time inflation ended at time $t_f = t_i + \Delta t$, the horizon must have expanded at least far enough to include today's observable universe to account for the uniformity of the CMB. We can therefore write the final horizon distance, at the end of inflation, as the present horizon distance, $d_{h,0} = 14.6$ Gpc (Eq. 29.159), multiplied by the scale factor $R(t_f)$. From Eq. (29.87),

$$d_f = d_{h,0} R(t_f) = (4.5 \times 10^{26} \text{ m})(2.04 \times 10^{-10} \text{ s}^{-1/2})(t_i + \Delta t_{\min})^{1/2}. \quad (30.15)$$

The minimum time interval consistent with the uniform CMB may then be found by solving

$$2ct_i e^{\Delta t_{\min}/\tau} = (4.5 \times 10^{26} \text{ m})(2.04 \times 10^{-10} \text{ s}^{-1/2})\sqrt{\tau} \left(\frac{t_i}{\tau} + \frac{\Delta t_{\min}}{\tau} \right)^{1/2}. \quad (30.16)$$

This may be solved numerically to obtain

$$\frac{\Delta t_{\min}}{\tau} = 62.4 \quad (30.17)$$

or

$$\Delta t_{\min} = 6.25 \times 10^{-35} \text{ s}. \quad (30.18)$$

During this brief time interval, the size of the observable universe grew from 6×10^{-28} m to 0.73 m! There is no reason to believe that the universe conspired to have inflation last exactly long enough to endow only the presently observable universe with a uniform CMB. It is customary to assume that the inflationary phase lasted for 10^{-34} s or more, during which the size of the universe increased by a factor of approximately $e^{100} \simeq 3 \times 10^{43}$. This is the roughest of estimates; the actual factor could have easily exceeded 10^{50} , a value that is also quoted.

Today's observable universe began inside a small bubble of true vacuum. We will take the initial diameter of the bubble to be the particle horizon distance at 10^{-36} s, just $d_i = 6 \times 10^{-28}$ m across. At the beginning of inflation, every point in the bubble lay within the particle horizon of every other point, and its contents were in thermodynamic equilibrium. The isotropy and homogeneity evident in our universe today were established at that early time. The bubble's final diameter, assuming inflation ended at 10^{-34} s, was

$$d_f = e^{100} d_i = 1.6 \times 10^{16} \text{ m.}$$

What fraction of this inflated bubble was occupied by the material that makes up the presently observable universe? To answer this question, we will work backward. Assuming a flat universe, the horizon distance to the farthest point that can be observed today is $d_{h,0} = 4.5 \times 10^{26}$ m (Eq. 29.159). According to Eq. (29.87), the scale factor at 10^{-34} s was $R = 4.9 \times 10^{-27}$. This means that the diameter of the presently observable universe at that time was, from Eq. (29.148),

$$2Rd_{h,0} = 4.4 \text{ m} = 2.8 \times 10^{-16} d_f.$$

This implies that the small bubble of true vacuum expanded to such an extent that when inflation ended at 10^{-34} s, today's observable universe occupied about 5 parts in 10^{46} of its volume. If the size of the presently observable universe immediately after inflation is represented by a hydrogen atom, then the inflated bubble of true vacuum in which it was immersed would be a sphere of radius 200 km!

This episode of inflationary growth came to a halt when spontaneous symmetry breaking finally brought an end to the GUTs epoch, and the strong nuclear force became distinct from the electroweak force. The elevated energy density of the false vacuum was then released, like the release of latent heat that occurs in freezing. This energy reheated the universe to nearly 10^{28} K, its preinflation value, and generated a burst of particle–antiparticle creation. From this point onward, the universe developed as previously described in the standard Big Bang picture. As shown in Fig. 30.4, the only difference lies in the first 10^{-34} s or so. But in a single stroke, this brief instant of inflation resolves the problems with the standard Big Bang.

Solutions to the Problems of the Standard Big Bang Theory

The exponential growth of our small bubble of true vacuum carried most of its volume far beyond the boundaries of today's universe. Nevertheless, because the bubble's volume was in thermodynamic equilibrium before inflation, the spectrum of the CMB is extremely smooth, thereby solving the horizon problem.

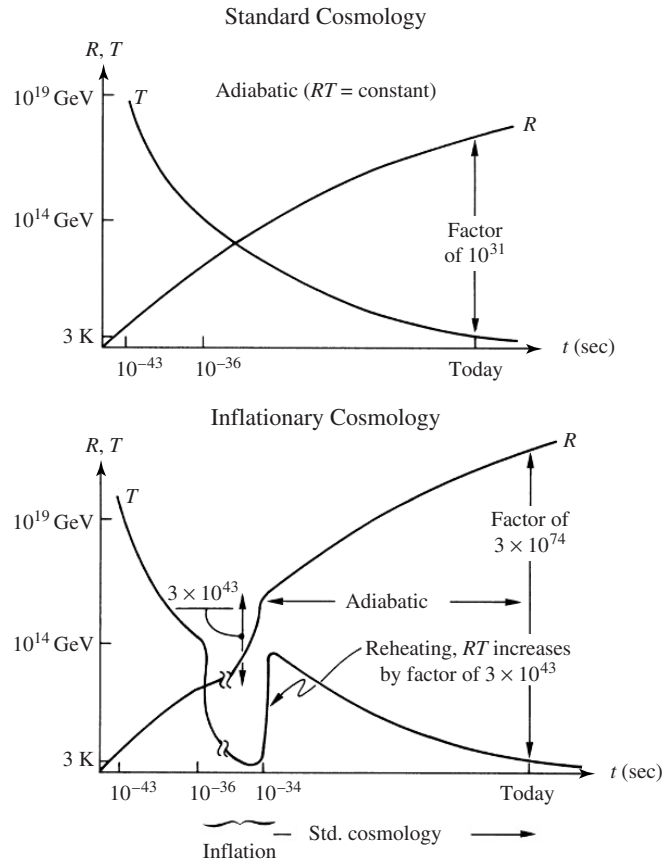


FIGURE 30.4 The evolution of the temperature of the universe and the scale factor, without and with inflation. Except for the bottom value, the temperature is given in terms of kT . [Figure adapted from Edward W. Kolb and Michael S. Turner, *The Early Universe* (page 274), ©1990 by Addison-Wesley Publishing Company, Inc., Reading, MA. Reprinted by permission of the publisher.]

Inflation also resolves the flatness problem. As the small bubble of true vacuum inflated, the scale factor R increased by a factor of e^{100} or more. Equations (29.194) and (30.12) show that during the episode of inflation, the density parameter would have varied according to

$$\Omega(t) = 1 + \frac{kc^2}{(dR/dt)^2} = 1 + \frac{kc^2\tau_i^2}{R^2(t)}.$$

The dramatic increase in R by at least 43 orders of magnitude forced $\Omega \rightarrow 1$. In the inflation model, the density parameter and curvature after inflation may be safely specified as $\Omega = 1$ and $k = 0$; the inflationary universe is *flat*. Note that this solution to the flatness problem arises naturally from the physics, instead of having to be specified as an ad hoc condition at the beginning.

The monopoles problem is also solved. Recall that at the end of the GUTs era, it is expected that defects arose at locations where a discontinuity in the choice of an internal

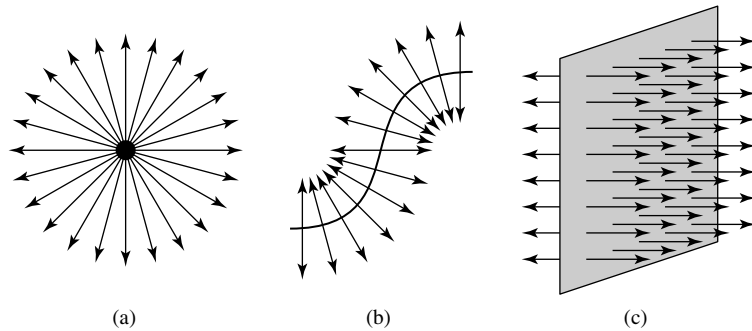


FIGURE 30.5 Types of defects: (a) magnetic monopoles, (b) cosmic strings, and (c) domain walls. The arrows illustrate the geometry of the discontinuity involved in each type of defect.

degree of freedom existed. These defects are remnants of the false vacuum, still in a state of unbroken symmetry with an energy density given by Eq. (30.9). Thus it is thought that they should be extremely massive. A pointlike discontinuity corresponds to a magnetic monopole, and it is conjectured that other discontinuities created additional relics. For example, a linear discontinuity gives rise to a **cosmic string**, whereas a planar discontinuity is called a **domain wall**. Figure 30.5 illustrates the geometry of these defects. A fourth type of defect called a **texture** is not shown, since the discontinuity is not localized to a point, line, or plane; it is more like a diffuse three-dimensional clump.

Monopoles and domain walls are so massive that they would easily overwhelm the gravitational effects of ordinary matter if they had not been diluted to insignificance by the inflation of the universe. These defects probably arose at the boundaries between homogeneous regions—that is, near the surfaces of the bubbles of true vacuum. It is thought that each bubble contained only a few monopoles or domain walls. As seen above, when the episode of inflation ended, the presently observable universe occupied an insignificant fraction of its bubble's volume (a few parts in 10^{46}), so it is expected that magnetic monopoles and domain walls should be exceedingly rare in today's universe.

Cosmic strings and textures could be more common in the present universe, however. A typical cosmic string, a long crack in space filled with false vacuum, might have a linear mass density of $\sim 10^{21} \text{ kg m}^{-1}$. In the simplest theories, either these thin tubes of false vacuum are infinitely long, or they form closed loops that are conjectured to be hundreds or thousands of parsecs across. A cosmic string might be detected through its influence as a gravitational lens, bending the starlight that passes by on either side to form a double image of the source. Tangled strings would curl and twist as they slowly straighten, forming loops that would sometimes cut the strings up into smaller loops. It is thought that these small loops eventually decay by emitting gravitational waves.

Matter–Antimatter Asymmetry

Another challenge cosmologists face is explaining why the universe consists of matter rather than antimatter. All but about 0.01% of cosmic rays which sample our Galaxy are matter rather than antimatter. The few antiparticles that are detected can be explained as the result of high-energy interactions, such as the creation of proton–antiproton pairs when two energetic protons collide. Rare ultra-high-energy cosmic rays (UHECR), with energies

exceeding 10^{11} GeV, may (or may not) originate in M87 in the Virgo cluster. (The source is only suggested, and the trajectory of UHECRs is complicated by the Milky Way's magnetic field.) Observations of UHECRs reveal no evidence of antimatter. Neither is there any sign, in the form of gamma rays of a certain energy, of the violent annihilation reactions that would occur if substantial concentrations of matter and antimatter were to collide. This indicates that within the region of space extending at least as far as the Galactic halo, and perhaps as far as the Virgo cluster, some 16 Mpc away, the universe is overwhelmingly composed of matter.

The explanation of this **matter–antimatter asymmetry** comes from a combination of the details of grand unified theories and the cooling of an expanding universe following inflation. As we have seen, any particles that were present before inflation would have been diluted to insignificance by the exponential expansion. All of the particles in our universe today originated in the burst of particle–antiparticle production that was fueled by the energy (latent heat) released by the false vacuum.⁸ The universe was filled with a soup of quarks, leptons, photons, and even more exotic hypothetical particles simply denoted X bosons, and their antiparticles $\bar{X}$. The spontaneous symmetry breaking that ended the GUTs epoch endowed the X particles with mass, just as the vector gauge bosons gained mass when the electroweak unification ended. These extremely massive X particles ($m_X c^2 \sim 10^{15}$ GeV) are not present at the much lower energies that characterize the universe today.

According to the grand unified theories, the X and $\bar{X}$ particles were present in equal numbers and could be transformed into pairs of quarks and antiquarks, respectively, according to

$$X \Rightarrow q + q \quad (30.19)$$

$$\bar{X} \Rightarrow \bar{q} + \bar{q}. \quad (30.20)$$

It is thought that the first of these reactions occurred slightly more frequently than the second.⁹

A related matter–antimatter asymmetry regarding reaction rates has been seen experimentally. One type of particle, called a *kaon*, can decay into another particle, called a *pion*, by the reactions $K \rightarrow \pi^- + e^+ + \nu_e$ and $K \rightarrow \pi^+ + e^- + \bar{\nu}_e$. The first reaction occurs slightly (but measurably) more frequently than the second.

Initially the temperature was so high that reactions (30.19) and (30.20) occurred equally in both directions. But as the universe cooled and the characteristic particle energy dropped, there was not enough energy available to replace the X s and $\bar{X}$ s as fast as they decayed. A permanent excess of quarks over antiquarks developed that survived the electroweak symmetry breaking at $t \sim 10^{-11}$ s and lasted until the universe was a few microseconds old. The universe was then cool enough ($T \sim 2 \times 10^{12}$ K) for the quark–hadron transition, when quarks and antiquarks combined to produce huge numbers of baryons and antibaryons, with a slight excess of baryons. Following this, there was a firestorm of particle–antiparticle annihilation that eliminated practically all of the antimatter, leaving only the small excess

⁸Alan Guth has referred to this as the “ultimate free lunch.”

⁹If you have some acquaintance with modern physics, you will recognize violations of both baryon number and CP (charge and parity) symmetry here. Both violations are integral features of grand unified theories.

of baryons that constitutes the visible matter in the universe today. The barrage of photons that was unleashed has since been cooled by the expansion of the universe to become the cosmic background radiation. Nearly all of the photons in the universe are from the CMB; the number produced by other sources (such as stars) pales in comparison. The ratio of the number of baryons in the universe to the number of photons is about 5×10^{-10} (see Problem 30.1). Because the annihilation of a baryon and an antibaryon produces two photons, this ratio implies that there was roughly one unpaired baryon for every one billion baryon–antibaryon pairs. These unpaired baryons were the tiny residue of matter that survived annihilation to make up the material world.

The CMB and the Decoupling of Matter and Radiation

For several thousand years following the formation of helium nuclei, the universe remained a hot broth of photons, hydrogen and helium nuclei, and electrons. Radiation dominated the dynamics of the expansion, and the scale factor of the universe increased as $R \propto t^{1/2}$. Finally, at $t \simeq 2 \times 10^{12}$ s, the CMB had become sufficiently diluted that massive particles began to govern the expansion. The radiation era came to a close and the matter era began at a temperature of roughly 9000 K.

The prevalence of free electrons continued to obstruct the CMB photons until the temperature fell to about 3000 K, when electrons combined with the nuclei to form neutral atoms. The resulting drop in the opacity decoupled the radiation from the matter in the universe and allowed the photons to roam freely. As we shall see, this decoupling had dramatic implications for the collapse of higher-density regions and the subsequent formation of structure in the universe.

30.2 ■ THE ORIGIN OF STRUCTURE

Even a momentary glimpse at the night sky provides convincing evidence that the universe is not perfectly homogeneous. Structure abounds on all scales, from planets to superclusters of galaxies. A typical galaxy is about 10^5 times more dense than the universe as a whole, so the present universe on a small scale is rather lumpy. This final section of the text deals with the emergence of structure from the featureless cauldron of the Big Bang, with galaxies and clusters of galaxies revealing the presence of early density fluctuations.

Adiabatic and Isothermal Density Fluctuations

The early universe (before recombination) was filled with a hot plasma of photons and particles. The electrons and photons were tightly coupled by Compton scattering and electromagnetic interactions, and the electrons coupled the protons to the photons. As a result, the photons, electrons, and protons moved together to form a **photon–baryon fluid**. Heisenberg’s uncertainty principle guarantees that, because the positions of particles are indeterminate, there must have been density inhomogeneities in the early universe. These disturbances were nearly the same on all scales, meaning that the density fluctuations caused almost the same magnitude perturbation of the gravitational potential. We say that the

density fluctuations were nearly **scale-invariant**.¹⁰ Their masses are thought to have ranged anywhere from the mass of stars to the mass of clusters of galaxies, with the less massive fluctuations being more common.

There are two types of density fluctuations. In principle, there can be independent fluctuations of the four components of the early universe: photons, baryons, neutrinos, and dark matter. If the fluctuations are proportional to one another in such a way that there is no energy exchange between them, they are called **adiabatic fluctuations** or **curvature fluctuations** (because the nonzero density fluctuation affects the local curvature of space-time). On the other hand, for **isothermal fluctuations** or **isocurvature fluctuations**, the sum of the four density fluctuations is zero. They involve a *potential* density increase that is suppressed or “frozen in.” For example, a spatial variation in the types of particles present may potentially produce a pressure difference, but this could not be converted into an actual density variation while particle motion was inhibited by the close interaction of matter with the uniform radiation field. Any collection of density fluctuations can be expressed as the sum of adiabatic fluctuations and isothermal fluctuations.

The distinction between adiabatic and isothermal fluctuations continued until decoupling, when radiation and matter went their separate ways. Freed from the drag of the surrounding radiation, particles in isothermal fluctuations could respond to differences in pressure and move to produce true density perturbations. The distinction between adiabatic and isothermal fluctuations then vanished. The importance of isothermal fluctuations lies in their ability to act as a deep freeze, preserving potential density fluctuations and protecting them from the dissipation process suffered by the adiabatic fluctuations (described later). Because nothing significant happened to the isothermal fluctuations until decoupling, we will concentrate on the story of their adiabatic counterparts.

The Development of Adiabatic Density Fluctuations

Quantum mechanical density fluctuations guarantee that, even in its first instant, the Big Bang was not perfectly smooth. Everywhere the density of regions of space vacillated randomly between being underdense and overdense. However, the baryonic matter, coupled to the photons, remained evenly distributed throughout the plasma. At about 10^{-36} s, inflation suddenly stretched the universe at an exponential rate so rapid that the size of the overdense and underdense regions vastly exceeded the particle horizon at that time. These density fluctuations were essentially frozen because they were not causally connected. The motion of their particles was inhibited because they could not respond to conditions beyond their horizon. The region therefore passively participated in the expansion of the universe.

To understand how a super-horizon-sized region of enhanced density evolved with time as the universe expanded, we will concentrate on a particularly simple case and imagine that, within an otherwise flat universe of density $\rho(t)$, there was a spherical region with an increased density of $\rho'(t) > \rho(t)$; see Fig. 30.6. Because the dynamics of the spherical density fluctuation depended only on the mass contained within it (recall Birkhoff’s theorem of Section 29.1), it evolved as a separate entity, like a miniature closed universe. Assuming

¹⁰We say *nearly* scale-invariant because inflation predicts that the fluctuations were a bit weaker on smaller mass scales. WMAP observations confirming this prediction were announced in March 2006, supporting the inflationary model.

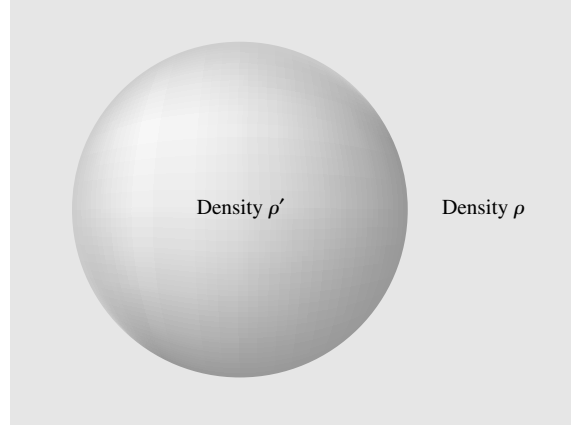


FIGURE 30.6 A spherical region of enhanced density ρ' , in a flat universe of density ρ .

that the Hubble flow has always been uniform, the region of higher density expanded at the same rate as its surroundings, so both regions can be described by the same scale factor, $R(t)$, and the same $H(t)$.¹¹ Equation (29.9) then shows that

$$H^2 R^2 - \frac{8}{3}\pi G \rho' R^2 = -kc^2$$

for the high-density region, and

$$H^2 R^2 - \frac{8}{3}\pi G \rho R^2 = 0$$

for the flat universe outside. Subtracting these results and dividing by $8\pi G \rho R^2$ shows that the density fluctuation $\delta\rho/\rho$ is

$$\frac{\delta\rho}{\rho} = \frac{\rho' - \rho}{\rho} = \frac{3kc^2}{8\pi G \rho R^2}. \quad (30.21)$$

During the radiation era, $\rho = \rho_{\text{rad}} \propto R^{-4}$ (Eq. 29.57) and the scale factor varied as $R \propto t^{1/2}$ (Eq. 29.86). Applying these dependencies to Eq. (30.21), we find that the density fluctuation is described by

$$\frac{\delta\rho}{\rho} = \left(\frac{\delta\rho}{\rho} \right)_i \left(\frac{t}{t_i} \right) \quad (\text{radiation era}), \quad (30.22)$$

where $(\delta\rho/\rho)_i$ is the value $\delta\rho/\rho$ at some initial time, t_i . The amplitude of our super-horizon-sized adiabatic fluctuation therefore increased linearly with time during the radiation era. (An isothermal fluctuation remained frozen with $\delta\rho/\rho = 0$.)

¹¹There is actually a slight difference in the scale factors, but for the early universe this can be neglected.

Similarly, during the matter era, $\rho \propto R^{-3}$ (Eq. 29.5) and the scale factor varied as $R \propto t^{2/3}$ for a flat universe (Eq. 29.31), so

$$\frac{\delta\rho}{\rho} = \left(\frac{\delta\rho}{\rho}\right)_i \left(\frac{t}{t_i}\right)^{2/3} \quad (\text{matter era}). \quad (30.23)$$

Note that the increase in $\delta\rho/\rho$ did not involve separation of the high-density region from the Hubble flow. The increase in the density perturbation was due to the difference in the rates at which the density decreased inside and outside the fluctuation as the universe expanded.

We now move ahead to a later time in the radiation era, when the particle horizon had expanded to encompass the entire adiabatic density fluctuation. The region of enhanced density is now causally connected and can respond as a whole to the physical conditions inside. From this point onward, the fate of the fluctuation is determined by the relative values of its mass and the Jeans mass.

The evolution of the Jeans mass as the universe expanded is of primary importance to our story. From Eq. (12.14) for a static (not expanding) medium, the minimum mass required for $\delta\rho/\rho$ to increase with time is¹²

$$M_J \simeq \left(\frac{5kT}{G\mu m_H}\right)^{3/2} \left(\frac{3}{4\pi\rho}\right)^{1/2}. \quad (30.24)$$

The same expression for the Jeans mass is valid for an expanding universe, although the consequences of falling below this minimum mass are different, as will be explained shortly.

The Jeans mass can be written in terms of the adiabatic sound speed. In general, the speed of sound v_s is given by

$$v_s = \sqrt{\frac{\partial P}{\partial \rho}}. \quad (30.25)$$

For the familiar adiabatic sound waves in air, $P = C\rho^\gamma$, where C is a constant, and we recover Eq. (10.84),

$$v_s = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{5kT}{3\mu m_H}}, \quad (30.26)$$

where the last expression comes from the ideal gas law, Eq. (10.11), with $\gamma = 5/3$ for an ideal monatomic gas. Rearranging and substituting into Eq. (30.24), the Jeans mass becomes

$$M_J = \frac{9\rho}{2\pi^{1/2}} \frac{v_s^3}{(G\rho)^{3/2}}. \quad (30.27)$$

The temperature dependence of the Jeans mass changed dramatically at the time of recombination. Although it is not obvious from Eq. (30.27), a more careful derivation of the Jeans

¹²From the derivation of the Jeans mass in Section 12.2, it should be noted that M_J must be in the form of baryonic matter. For example, the use of $K = 3/2 NkT$ in the derivation of M_J may not be applied to nonbaryonic particles.

mass applicable to the time before recombination reveals that the ρ in the numerator is ρ_b , the baryonic mass density. Because the temperatures of radiation and matter were equal during this epoch, Eqs. (29.5) and (29.58) show that $\rho_b = \rho_{b,0}/R^3 = \rho_{b,0}T^3/T_0^3$. However, the ρ in the denominator of Eq. (30.27) includes the dominant effect of the photons, so $\rho \propto T^4$ in the denominator (Eq. 29.77). Furthermore, before recombination $P_{\text{rel}} = u_{\text{rel}}/3 = \rho_{\text{rel}}c^2/3$ (Eq. 9.11), so the speed of sound was

$$v_s = \sqrt{\frac{\partial P}{\partial \rho}} = \frac{c}{\sqrt{3}}, \quad (30.28)$$

about 58 percent of the speed of light. The result, as shown in Fig. 30.7, is that the Jeans mass was proportional to T^{-3} until the time of recombination approached. After recombination, the density in the denominator reverted to a mass density, and the sound speed plunged to assume its ideal gas value. Because radiation and matter are decoupled after recombination, the mass density is independent of the temperature. The only temperature dependence is that of the sound speed, Eq. (30.26), and so after recombination the Jeans mass is proportional to $T^{3/2}$. Figure 30.7 shows the temperature dependence of the Jeans mass. In reality, the presence of dark matter levels off the sharp peak before the time of recombination to produce a plateau at about $M_J \simeq 10^{16} M_\odot$.

Before recombination occurred, the Jeans mass was about an order of magnitude *greater* than the amount of baryonic matter contained within a causally connected region (see Problem 30.7), so the mass of a sub-horizon-sized density fluctuation was always less than the Jeans mass. Consequently, during this period the growth of an adiabatic density fluctuation was suppressed; in other words, $\delta\rho/\rho$ remained constant until the passage to the matter era.

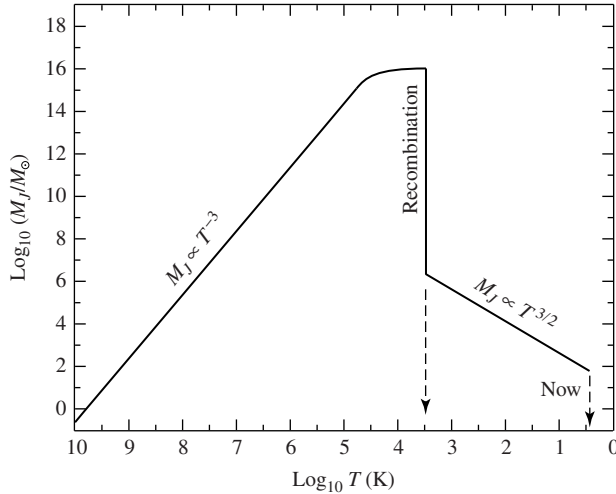


FIGURE 30.7 The variation of the Jeans mass, M_J , with temperature as the universe expands. The sharp peak at the time of recombination is actually leveled off by the presence of dark matter at about $M_J \simeq 10^{16} M_\odot$.

Acoustic Oscillations and Damping

From the time an adiabatic fluctuation became sub-horizon-sized until recombination, it underwent acoustic oscillations reminiscent of stellar pulsation (see Section 14.2). Sound waves traversed its volume at a substantial fraction of the speed of light ($c/\sqrt{3}$) until recombination occurred. These sound waves produced regions of compression (slightly hotter) and rarefaction (slightly cooler) that left their imprint on the CMB when decoupling occurred. We will return to these immensely useful acoustic oscillations after our discussion of the fate of the density fluctuations.

The smaller adiabatic fluctuations did not survive the phase of acoustic oscillations, since photons could diffuse and leak out of compressed regions when the wavelengths of the sound waves were short enough. The close interaction between the escaping radiation and matter then produced a smoothing effect that damped out the short-wavelength compressions.

The Minimum Mass Required to Survive Damping

To determine how large a fluctuation needed to be to survive this damping, consider a region of enhanced density just prior to the time of decoupling, at $t_{\text{dec}} = 3.79 \times 10^5$ yr (Eq. 29.102). As a rough estimate, the density fluctuation would survive if the size of the region were larger than the distance that a diffusing, random-walking photon could travel during the age of the universe up to decoupling. The mean free path of the photons is given by Eq. (9.12), $\ell = 1/n\sigma$, where $\sigma = 6.65 \times 10^{-29}$ m² is the electron-scattering cross section (Eq. 9.20) and n is the electron number density. The number of scatterings suffered by a photon in the age of the universe is then

$$N = \frac{t_{\text{dec}}}{\ell/c} = n\sigma ct_{\text{dec}}.$$

The average displacement from the photon's starting point can be obtained from Eq. (9.29),

$$d = \ell\sqrt{N} = \frac{\sqrt{n\sigma ct_{\text{dec}}}}{n\sigma} = \sqrt{\frac{ct_{\text{dec}}}{n\sigma}}.$$

A sound wave with a wavelength longer than d could survive the damping effect of the diffusing photons. The damping of the shorter-wavelength oscillations is known as **Silk damping**, named for American astronomer Joseph Silk.

The minimum mass of a surviving density fluctuation must have been roughly the mass contained within a sphere of radius d . Equations (29.5) and (29.4) show that the baryonic mass density at the time of decoupling (when $z = 1089$) was

$$\rho_b = \frac{\rho_{b,0}}{R^3} = \rho_{b,0}(1+z)^3 \simeq 5.4 \times 10^{-19} \text{ kg m}^{-3}, \quad (30.29)$$

where the present baryonic mass density, $\rho_{b,0}$, is given by (Eq. 29.17). The number density of electrons can now be found from $n \simeq \rho_b/m_H$, assuming, for convenience, composition of pure hydrogen (one electron for every proton). Setting $\rho = \rho_b$, the mass contained in

this spherical volume was then about

$$\frac{4}{3}\pi d^3 \rho_b = \frac{4\pi}{3\rho_b^{1/2}} \left(\frac{m_H c t_{\text{dec}}}{\sigma} \right)^{3/2} = 7.7 \times 10^{13} M_\odot.$$

Any adiabatic fluctuations with a mass smaller than this did not survive past recombination. It is interesting that this value of several $\times 10^{13} M_\odot$ is reminiscent of the mass of an elliptical cD galaxy or a small cluster of galaxies.

The Isothermal Density Fluctuations

We now return our attention to the isothermal density fluctuations. Density perturbations less massive than some $10^{13} M_\odot$ existed after recombination only because the “frozen” isothermal fluctuations were preserved from the dissipative effects of the acoustic oscillation phase. After recombination, these fluctuations were transformed into actual regions of enhanced density. The pressure perturbations, formerly restrained by the radiation field, were freed to move particles and produce true density differences. Consequently, there is no further need to distinguish between adiabatic and isothermal density fluctuations.

The Jeans Mass after Recombination

The deviations from the average density were fairly small at the time of recombination ($\delta\rho/\rho \sim 10^{-4}$, as will be seen later), but the fluctuations encompassed various amounts of mass, presumably ranging from stellar to galactic cluster values. We can evaluate the Jeans mass just after recombination, when $T \simeq 2970$ K and the density is given by Eq. (30.29). With a mean molecular weight of $\mu = 0.584$ for an assumed composition of $X = 0.77$ and $Y = 0.23$ (see Eq. 10.16), the value of the Jeans mass was

$$M_J \simeq \left(\frac{5kT}{G\mu m_H} \right)^{3/2} \left(\frac{3}{4\pi\rho} \right)^{1/2} \simeq 1.9 \times 10^6 M_\odot,$$

comparable to the mass of a globular cluster. After recombination, fluctuations with masses exceeding about $10^6 M_\odot$ were amplified. The rate of growth of the density perturbations is just that given by Eq. (30.23), $\delta\rho/\rho = (\delta\rho/\rho)_i (t/t_i)^{2/3}$.

The Timing of Structure Formation

It is remarkable that the two characteristic mass values that have emerged from our analysis of density fluctuations, $10^6 M_\odot$ and $10^{13} M_\odot$, neatly span a range of masses that includes most of the galactic objects of interest, from stellar clusters to clusters of galaxies. We will now follow the evolution of these higher-density regions and investigate whether they could have produced the objects that are actually observed in the universe.

Example 30.2.1. The highest-redshift quasar is J114816.64+525150.3 (as of this writing). Discovered by the Sloan Digital Sky Survey, its redshift is $z = 6.43$, corresponding to $R_q = 1/(1+z) = 0.135$ (the q subscript denotes “quasar”). Using Eq. (29.129) reveals

continued

that the age of the universe when the light left the quasar is, using WMAP values, $2.75 \times 10^{16} \text{ s} = 872 \text{ Myr}$. We will assume that the gravitational collapse of a region can begin when $\delta\rho/\rho \approx 1$. At this point, nonlinear effects become important, and the region separates from the Hubble flow and starts to collapse. Setting $(\delta\rho/\rho)_q = 1$ for the collapse that formed this farthest quasar, Eq. (30.23) can be used to find the size of the density fluctuation at the time of recombination, $[t_{\text{dec}}]_{\text{WMAP}} = 3.79 \times 10^5 \text{ yr}$ (see Eq. 29.102). The result is

$$\left(\frac{\delta\rho}{\rho}\right)_{\text{rec}} = \left(\frac{\delta\rho}{\rho}\right)_q \left(\frac{t_{\text{dec}}}{t_q}\right)^{2/3} \simeq 0.005.$$

To manufacture the most distant quasars, there must have been density fluctuations of several tenths of a percent at the time decoupling occurred.

The results of this example are contradicted by careful observations of the anisotropies in the cosmic background radiation. At the time of decoupling, photons would have experienced a gravitational redshift as they climbed out of the gravitational potential wells that surrounded the largest fluctuations. (These largest fluctuations are discussed further on page 1268.) The radiation from regions of enhanced density should appear slightly cooler than the background CMB, while radiation from lower-density regions should appear a bit hotter. These fluctuations in the temperature of the CMB have been measured to be in the range of $\delta T/T \simeq 1\text{--}7.5 \times 10^{-5}$. These properties can be translated into density variations by using Eqs. (29.5) and (29.58), which together imply that $\rho \propto T^3$. It is left as an exercise to show that the linearized version of this relationship is

$$\frac{\delta\rho}{\rho} = 3 \frac{\delta T}{T}. \quad (30.30)$$

The resulting values of $\delta\rho/\rho \simeq 0.3\text{--}2.3 \times 10^{-4}$ are about two orders of magnitude less than those predicted from arguments like the one made in Example 30.2.1. The accumulations of baryonic matter represented by these temperature fluctuations could not have grown fast enough by themselves to collapse and form the large-scale structures that are common in the present universe. There must have been density fluctuations hundreds of times greater than these at the time of recombination, even though this simple analysis of baryonic matter does not detect them.

Nonbaryonic dark matter may provide the solution to this puzzle. If this dark matter has a negligible interaction with radiation, it could have started to accumulate long before decoupling occurred, at the beginning of the matter era.¹³ (Recall that baryonic fluctuations could not grow during this time because the close interaction of radiation and baryonic matter resulted in a sound speed of $c/\sqrt{3}$, which in turn produced a prohibitively high value of the baryonic Jeans mass.) During the matter era, these concentrations of dark matter would have grown at the rate given by Eq. (30.23), $\delta\rho/\rho = (\delta\rho/\rho)_i (t/t_i)^{2/3}$. By the time of decoupling, the relative density enhancement of the dark matter could have reached the level of several tenths of a percent, as required by Example 30.2.1. After decoupling,

¹³During the earlier radiation era, the universe was expanding too fast to allow any sub-horizon-sized density fluctuations to grow, regardless of their composition.

TABLE 30.3 Redshifts for Structure Formation. Approximate redshifts at the time of the formation of various structures. (Adapted from Peebles, *Principles of Physical Cosmology*, Princeton University Press, Princeton, NJ, 1993.)

Structure	Redshift
Density fluctuations in CMB	$[z_{\text{dec}}]_{\text{WMAP}} = 1089$
Spheroids of galaxies	$z \sim 20$
The first engines of active galactic nuclei	$z \geq 10$
The intergalactic medium	$z \sim 10$
Dark halos of galaxies	$z \sim 5$
The first 10% of heavy elements	$z \geq 3$
Rich clusters of galaxies	$z \sim 2$
Thin disks of spiral galaxies	$z \sim 1$
Superclusters, walls, and voids	$z \sim 1$

baryonic matter would have been gravitationally attracted to these clumps of dark matter. It is thought that the sizes of the resulting baryonic fluctuations would quickly catch up with those of the underlying dark matter and that thereafter they would share the same value of $\delta\rho/\rho$ as they grew to form the concentrations of ordinary matter that are visible throughout the universe. When $\delta\rho/\rho \approx 1$, these regions of dark and baryonic matter would detach from the Hubble flow and begin their collapse to form the spheroidal components of galaxies, the central engines of quasars, and other early-forming structures; see Table 30.3. However, the time constraints are severe, and it is still not clear whether the collapse of regions of dark matter can account for the earliest observed structures.

Determining When the First Stars and Galaxies Formed

A galaxy discovered far beyond the galactic cluster Abell 2218 illustrates just how tight the timing for galaxy formation is. With a redshift of approximately $z \simeq 7$, this is the most remote galaxy found so far. The light we observe left this galaxy when the universe was just $1/(1+z) = 0.125$ of its present size. Using Eq. (29.129) with WMAP values, this corresponds to a time just 780 million years after the Big Bang, just 5.7% of the present age of the universe. The galaxy is tiny, perhaps just 600 pc in diameter. Nevertheless, it is a site of extremely active star formation. It is fortunate that Abell 2218 served as a gravitational lens that magnified the image of this farthest galaxy and allowed it to be detected. The presence of such a young galaxy serves as a severe constraint on theories of structure formation. Because it formed less than 1 billion years after the Big Bang, it must have formed extremely rapidly (perhaps by the process of dissipative collapse discussed in Section 26.2).

One clue pointing to the time when the first stars and galaxies were formed comes from observations of the Lyman- α forest in high- z quasars (recall Section 28.4). The closely spaced Ly α absorption lines are produced by clouds of neutral hydrogen of smaller redshift that lie between the quasar and Earth. Neutral hydrogen is a very efficient absorber of 122-nm (UV) photons. However, if, after recombination at $z = 1089$, all of the hydrogen in the intergalactic medium remained neutral, then almost all of the Ly α forest should be reduced to

zero due to absorption by the intergalactic neutral hydrogen. In effect, the neutral hydrogen in the intergalactic medium would act as an absorbing cloud with a continually declining redshift in the direction of Earth. This indicates that the hydrogen in the intergalactic medium is not neutral but almost completely ionized.

We conclude that after recombination, the universe entered a “Dark Age,” before the first stars and galaxies had formed and started to shine. Then UV radiation from the first generation of stars and AGN reionized the universe, and it has remained ionized until the present time. If we can find a flattening of the $\text{Ly}\alpha$ forest for a high- z quasar, then we know that we are observing that quasar during the **epoch of reionization**. This flat region in a quasar’s spectrum is called the **Gunn–Peterson trough**, named for the astronomers who predicted it, James Gunn (American) and Bruce Peterson (Australian). Figure 30.8 shows the Gunn–Peterson trough for four high- z quasars. The relatively rapid diminishing of the $\text{Ly}\alpha$ forest indicates that the reionization of the intergalactic neutral hydrogen was finished by $z = 6$ and that its final stages occurred very quickly. We may very well be penetrating back to the end of the “Dark Age.”

According to the WMAP results, the first stars ignited about 200 million years after the Big Bang. The Spitzer Space Telescope may have detected the light (shifted to infrared wavelengths) from the first generation of objects that initiated the reionization of the universe. Following a deep, 10-hour exposure in the direction of the constellation Draco, all known objects were carefully subtracted from the image. An infrared background remained, with blobs that may be the glow from the very first stars (Population III), as shown in Fig. 30.9.

Top-Down Galaxy Formation and Hot Dark Matter

As noted previously, it is intriguing that the two mass values to emerge from recombination roughly span the mass range of galactic structure. The Jeans mass of $M_J \simeq 10^6 M_\odot$ is typical of a globular cluster of stars. In addition, the lower limit for the mass of adiabatic fluctuations that survived the acoustic oscillation phase before recombination, $10^{13} M_\odot$, is characteristic of an elliptical cD galaxy or a small cluster of galaxies. (Adiabatic density fluctuations and hot dark matter are thought to behave similarly, because they both tend to resist clumping through the diffusion of photons and fast-moving particles.)

The resulting top-down process of galaxy formation involves the breakup of larger structures. The problem with this process is that the breakup may occur too late to be consistent with the observed times of formation of the earliest galaxies. Any clumping that occurs involves large amounts of mass (10^{13} to $10^{15} M_\odot$, representative of galactic clusters and superclusters) and so must be involved in a top-down process for forming galaxies from the breakup of larger entities.

Bottom-Up Galaxy Formation and Cold Dark Matter

On the other hand, cold dark matter is slow moving and should accumulate much more easily. Concentrations of cold dark matter can begin to collect on a variety of mass scales when the matter era begins, resulting in a bottom-up scenario for forming galaxies from the assembly of smaller components. The difficulty with the cold dark matter model is that numerical simulations of the early universe have trouble reproducing the galactic voids found in redshift surveys, such as those shown in Fig. 27.23. It has been found that if

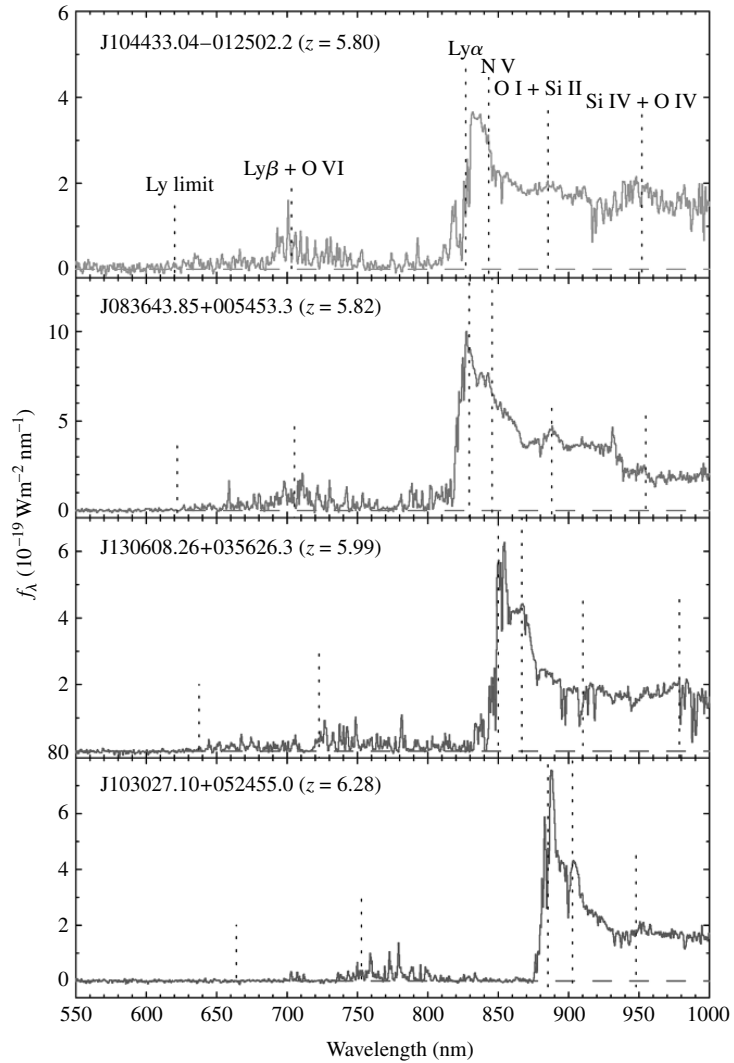


FIGURE 30.8 The Gunn–Peterson trough observed in four high- z quasars illustrates the rapid suppression of the Ly α forest with increasing z . This indicates that the abundance of ionized hydrogen has declined significantly from $z \sim 5$ to 6 and that the universe is approaching the reionization epoch at $z \sim 6$. (Figure adapted from Becker et al., *A. J.*, 122, 2850, 2001.)

galaxy formation is *biased*, so galaxies tend to form more readily in overdense regions, this problem with cold dark matter can be overcome. Figure 30.10 shows numerical simulations of the formation of structure from the bottom-up clumping of cold dark matter and from the top-down fragmentation of hot dark matter. As explained in Sections 24.2 and 26.2, current observations and interpretations strongly favor a bottom-up process for the formation of galaxies. The bright galaxies we observe today were assembled from fragments at high redshift.

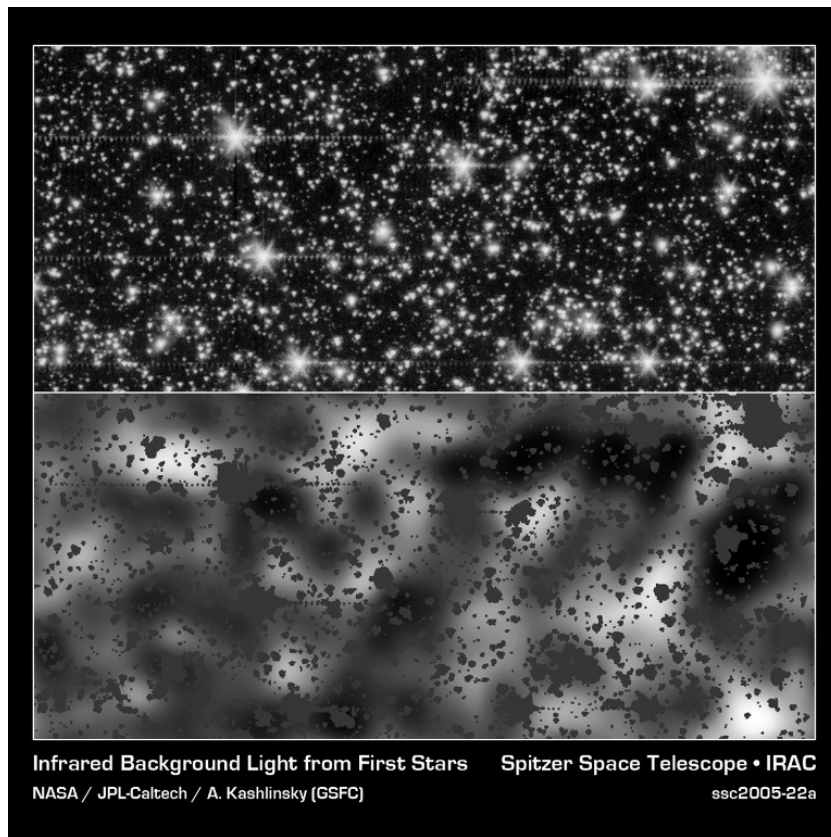


FIGURE 30.9 The top panel shows the results of a 10-hour exposure by the Spitzer Space Telescope. The lower panel shows the infrared emission, perhaps from the first generation of stars, that remains when all known objects have been subtracted.

The presence of high-redshift galaxies and quasars presents severe challenges for every theory of structure formation. The Sloan Digital Sky Survey has discovered several luminous quasars with $z > 6$ that indicate the existence of supermassive black holes of 10^9 to $10^{10} M_{\odot}$ when the universe was less than 1 Gyr old. It is significant that the most realistic numerical simulation of structure formation can reproduce these very early components of the universe. The Virgo consortium, an international group from the United Kingdom, Germany, Japan, Canada, and the United States, computed the behavior of $\sim 10^{10}$ particles as they evolved from $z = 127$ to form some 2×10^7 galaxies in a cube with sides of $500h^{-1}$ Mpc, a project the consortium called the Millennium Simulation. The results (see Fig. 30.11) demonstrate that a few massive black holes can form quickly enough (by a few 10^8 years) to account for the SDSS observations. These black holes evolve to become the most massive galaxies, which reside at the center of the largest clusters of galaxies.

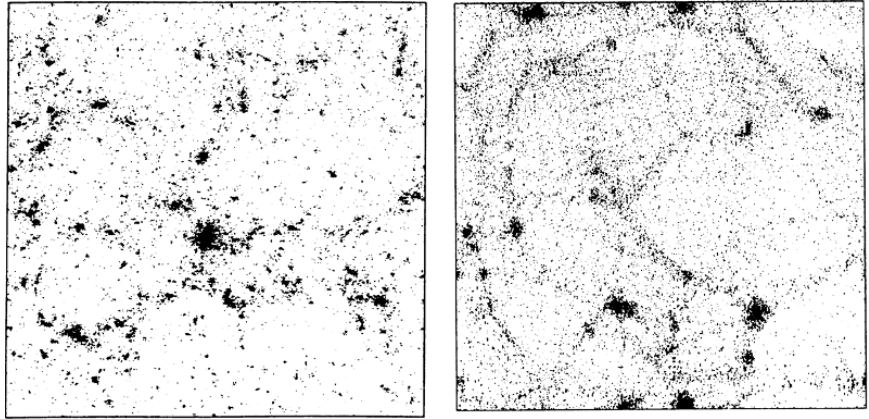


FIGURE 30.10 The results of numerical simulations of the formation of structure in the universe from the bottom-up clumping of cold dark matter (left), and the top-down fragmentation of hot dark matter (right). Both models assume $\Omega_0 = 1$ and $h = 0.5$. (Figure from Frenk, *Physica Scripta*, T36, 70, 1991.)

Cosmic Harmonics and Acoustic Oscillations

“The music of earth sings through a thousand holes.
The music of man is made on flutes and instruments.
What makes the music of heaven?”

— Chuang Tzu (4th century B.C.)¹⁴

We now return to the acoustic oscillations mentioned on page 1252. Recall that before recombination, the photons, electrons, and protons moved together to form a photon–baryon fluid. Quantum mechanical density fluctuations caused the density of regions of space to vary randomly between being underdense and overdense. At about 10^{-36} s, inflation suddenly stretched the universe at an exponential rate, so fast that the size of the overdense and underdense regions vastly exceeded the particle horizon at that time. These density fluctuations were essentially frozen; they could not respond as a whole because they were not causally connected.

When the particle horizon finally grew large enough to engulf the density fluctuations, the fluctuations were set free to react to their environment. At this time, when the universe was roughly 10^5 years old, the universe became filled with sound waves that traveled through the plasma. As the density of a region was enhanced by a compression, it was heated by a small amount ($\delta T/T \approx 10^{-5}$) until radiation pressure reversed the motion. Similarly, as the density was diminished by an expansion, it cooled. The small amplitude of the sound waves means that a passing wave will cause the photon–baryon fluid to execute simple harmonic motion. This motion continued until the time of decoupling, when the electrons combined with the protons and heavier nuclei formed in Big Bang nucleosynthesis. The

¹⁴Quotation from *The Way of Chuang Tzu*, Thomas Merton (trans.), New Directions, 1965.

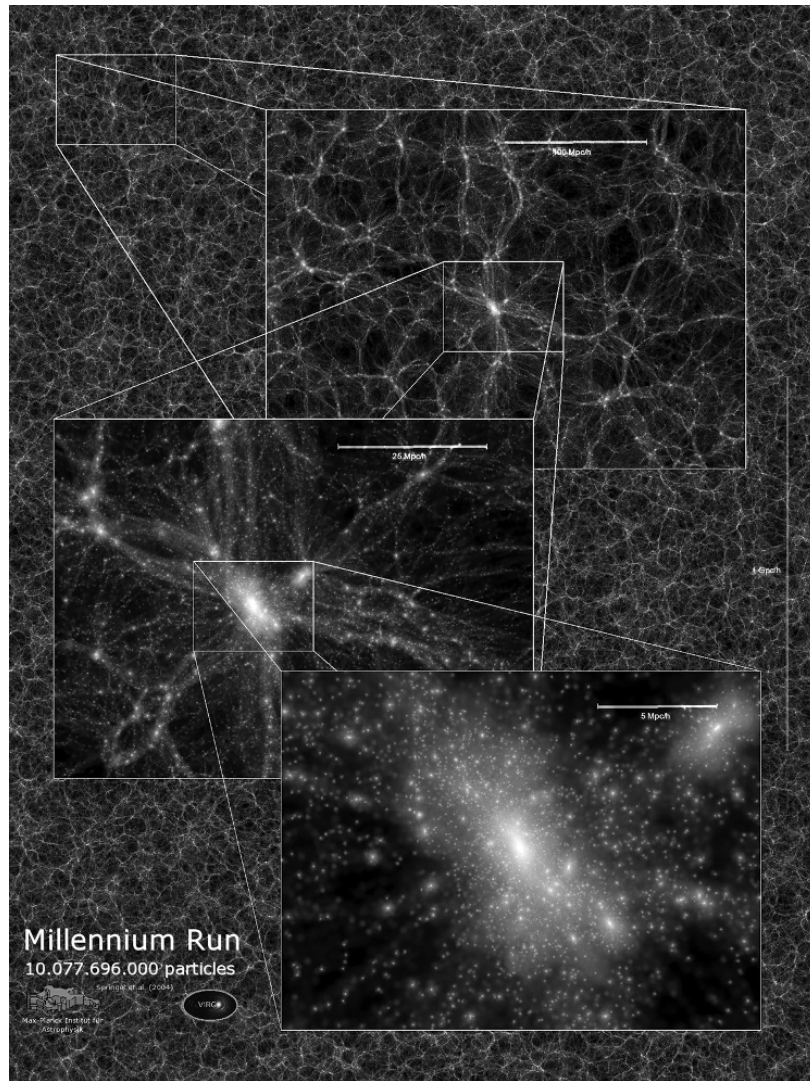


FIGURE 30.11 The Millennium Simulation of 200 million galaxies. This is a $15 \text{ Mpc } h^{-1}$ thick slice of the results at $z = 0$. The panels zoom in by a factor of four. The maximum magnification shows the substructure of a rich cluster of galaxies. (Figure courtesy of the Virgo consortium.)

photons of the CMB, now released to traverse the universe freely, carried the signature of the acoustic oscillations. Some areas of the CMB appear slightly hotter (higher frequency), others slightly cooler.

The frequency of the photons is also affected by the **Sachs–Wolfe effect** as the photons lose energy when they climb out of the gravitational potential wells of the density fluctuations. Overdense regions cause the photon frequency to be reduced slightly (hence appearing cooler), whereas underdense regions have the opposite effect.

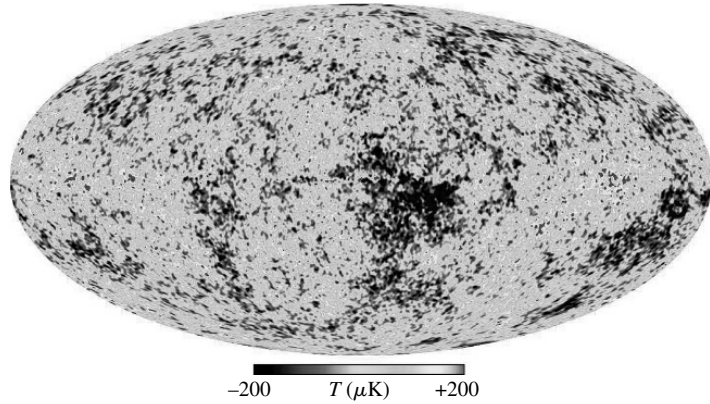


FIGURE 30.12 The temperature fluctuations in the CMB observed by the WMAP orbiting observatory. (Figure adapted from Bennett et al., *Ap. J. Suppl.*, 148, 1, 2003.)

The overall pattern of CMB temperature variations on the celestial sphere (see Fig. 30.12) can be expressed as the sum of spherical harmonic functions $Y_m^\ell(\theta, \phi)$, encountered previously in Chapter 14. Referring to Fig. 14.16, note that the pattern for a given $Y_m^\ell(\theta, \phi)$ has ℓ nodal lines [where $Y_m^\ell(\theta, \phi) = 0$], with $|m|$ of these lines passing through the figure's north pole ($\phi = 0$).

The temperature fluctuation in the direction of the angular position (θ, ϕ) is

$$\frac{\delta T(\theta, \phi)}{T} = \sum_{\ell=1}^{\infty} \sum_{m=-\ell}^{\ell} a_{\ell,m} Y_m^\ell(\theta, \phi).$$

The coefficients $a_{\ell,m}$ are in general complex numbers because the spherical harmonics themselves are complex. The value of the $a_{\ell,m}$ can be determined by observing the CMB and making measurements of $\delta T/T$ in all directions. The $\ell = 1$ is the dipole anisotropy of the CMB (the Doppler shift of the CMB caused by an observer's peculiar velocity through space relative to the Hubble flow; see Eq. 29.62), which is ignored in the following analysis.

To remove the effect of the arbitrary choice of the direction of $\phi = 0$, an angular average over the $2\ell + 1$ values of m can be taken:

$$C_\ell = \frac{1}{2\ell + 1} \sum_{m=-\ell}^{\ell} |a_{\ell,m}|^2 = \frac{1}{2\ell + 1} \sum_{m=-\ell}^{\ell} a_{\ell,m} a_{\ell,m}^*,$$

where a^* is the complex conjugate of a . The angular power spectrum is then defined to be $\ell(\ell + 1)C_\ell/2\pi$. Note that every term in the summation is ≥ 0 , so both positive and negative temperature fluctuations contribute to the angular power spectrum; they do not cancel each other.

The values of C_ℓ contain much information about the physical conditions and constituents of the early universe. Figure 30.13 shows the angular power spectrum of the CMB and the best-fit theoretical Λ CDM model. This is an amazing result. In musical terms, it is analogous to the waveform generated by displaying the sound of an instrument (a flute, say) on an

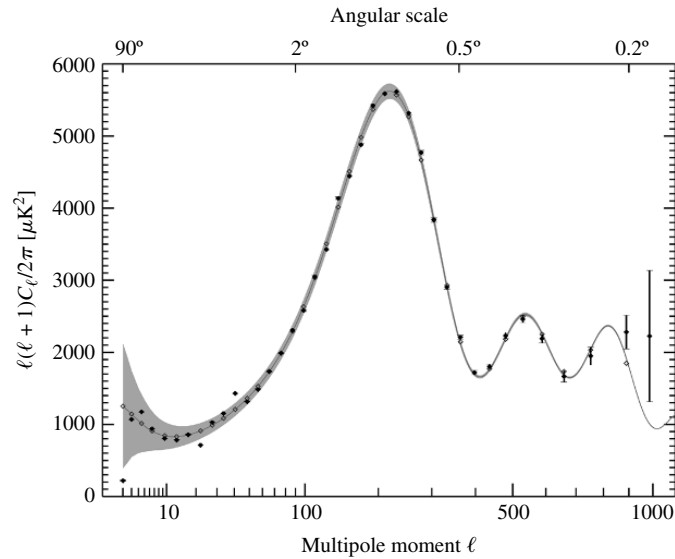


FIGURE 30.13 The angular power spectrum of the temperature fluctuations in the CMB. The solid line is the best-fit Λ CDM model. [Figure adapted from Hinshaw et al., *Ap. J.* (submitted), 2006, and courtesy of NASA and the WMAP Science Team.]

oscilloscope. The waveform identifies the fundamental and harmonic frequencies of the flute, and the height of the peaks reveals the relative power of the harmonics. Similarly, Fig. 30.13 shows the fundamental (the highest peak) and two harmonics of the acoustic oscillations of the early universe. Just as the air inside the flute has several modes of vibration, so does the CMB on the celestial sphere.

Fine-Tuning Our Understanding of Cosmic Harmonics

But why does the cosmic symphony involve these particular harmonics? Why is there a low, flat plain for small values of ℓ , a fundamental peak around $\ell \simeq 200$ (which corresponds to an angular size of about 1° on the sky), and then a few harmonics peaks whose heights decline as ℓ approaches 1000? These are difficult questions because we are actually looking at a cross section of the sound waves, where they intersect the surface of last scattering (a complication we will conveniently ignore). Nevertheless, we can use some simple models to understand the main features of the angular power spectrum.

Let's begin with the first peak, at $\ell \simeq 200$. We want to determine the angular diameter of the largest region that could have responded to a sound wave. For the region to respond as a whole, a sound wave must have been able to travel across the region by the time of decoupling t_{dec} . This is known as the **sonic horizon distance**. Because the speed of sound before decoupling is proportional to the speed of light, the sonic horizon distance d_s is related to the particle horizon distance d_h by $d_s = d_h/\sqrt{3}$. The time of radiation-matter equality is much earlier than the time of decoupling, $t_{r,m} \ll t_{\text{dec}}$ [see Eqs. (29.85) and (29.102)], so the acoustic oscillations occurred primarily during the matter era. Employing Eq. (29.155) for

the particle horizon distance during the matter era, $d_h = 3ct$, the sonic horizon distance is thus

$$d_s(t) = ct\sqrt{3}.$$

Using Eq. (29.102) for the time of decoupling, we obtain

$$d_s(t_{\text{dec}}) = ct_{\text{dec}}\sqrt{3} = 6.21 \times 10^{21} \text{ m} = 201 \text{ kpc}.$$

We now estimate the proper distance to the surface of last scattering, the origin of the CMB photons and the site of our region of diameter $d_s(t_{\text{dec}})$. For a flat universe, the proper distance d_p and angular distance d_A are related by [cf. Eqs. (29.169), (29.173), and (29.191)],

$$d_p = d_A(1+z) = \frac{D(1+z)}{\theta}, \quad (30.31)$$

where θ is the angular diameter of an object of physical diameter D located at redshift z . Using Eq. (29.145) and $R(t) = Ct^{2/3}$ for the matter era [Eq. (29.91), C is a constant], we obtain the proper distance

$$d_p = Ct_0^{2/3} \int_{t_{\text{dec}}}^{t_0} \frac{c dt}{Ct^{2/3}} = 3ct_0 \left[1 - \left(\frac{t_{\text{dec}}}{t_0} \right)^{1/3} \right] = \frac{D(1+z)}{\theta}. \quad (30.32)$$

Setting $D = d_s(t_{\text{dec}})$ and using $[z_{\text{dec}}]_{\text{WMAP}} = 1089$ along with the WMAP values of t_{dec} and t_0 , we find that

$$\theta = 1.79 \times 10^{-2} \text{ rad} = 1.03^\circ,$$

which corresponds to $\ell \simeq \pi/\theta = 175$, reasonably close to the observed value of $\ell \simeq 200$ for the first acoustic peak. A more detailed analysis shows that the location of the first peak is sensitive to the value of Ω_0 and is given by $\ell \simeq 200/\sqrt{\Omega_0}$. In effect, we are looking at a region of known size d_s through the gravitational lens of the overall geometry of the universe, as determined by Ω_0 , and comparing its angular size with our theoretical expectations. The Boomerang observatory,¹⁵ carried aloft by a stratospheric long-duration balloon, was one of the first to accomplish this measurement. Figure 30.14 illustrates the comparison of Boomerang's observations with theoretical simulations for a closed, a flat, and an open universe. The results convincingly demonstrate that the universe is indeed very nearly flat.

A Simple Model of Acoustic Oscillations

To investigate further the acoustic oscillations, we will consider a model problem that still retains the essential physics. Consider a cylinder of cross-sectional area A and length $2L$ that is filled with a gas; see Fig. 30.15. In order to model the ebb and flow of the photon–baryon

¹⁵The acronym stands for Balloon Observations Of Millimetric Extragalactic Radiation ANisotropy and Geophysics.

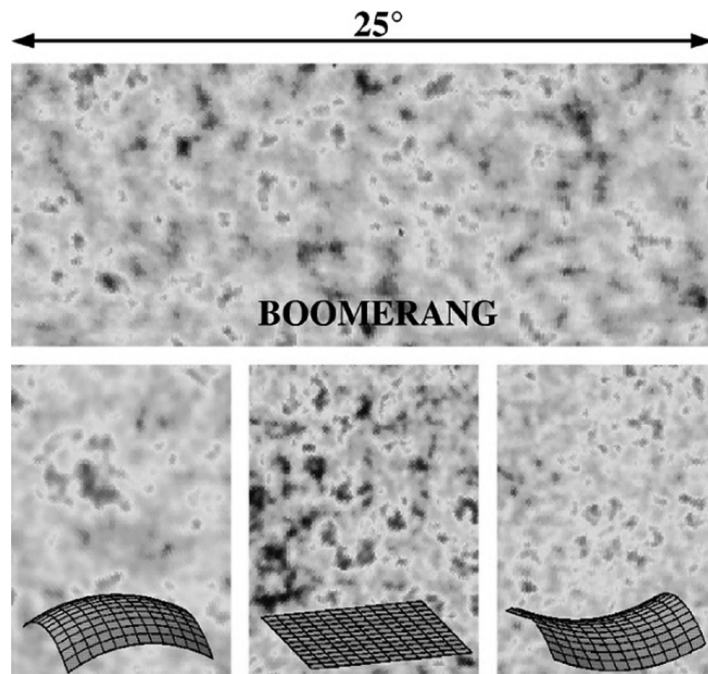


FIGURE 30.14 A comparison of Boomerang's observations (top) of temperature fluctuations in the CMB with theoretical simulations for a closed (left), flat (center), and open (right) universe. The results show that the universe is very nearly flat. (Image courtesy of the Boomerang Collaboration.)

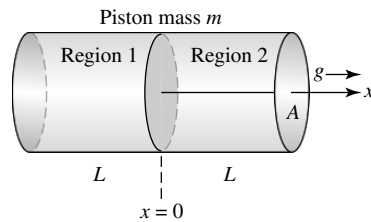


FIGURE 30.15 A cylinder of length $2L$ and cross-sectional area A filled with a gas. A movable piston is at the middle of the cylinder.

fluid as a sound wave passes through it, we want to study how the gas in the cylinder can experience simple harmonic motion. Of course the gas in the cylinder will form a standing sound wave, but we want to approximate the physics rather than write down the exact answer. To this end, let there be a movable piston in the middle of the cylinder. The piston represents the inertia of the baryons, so we will set the mass of the piston equal to the mass of the gas in the cylinder. If the equilibrium values of the pressure and density are P_0 and ρ_0 , respectively, then the mass of the piston is $m = 2LA\rho_0$.

If the piston is displaced, the density on either side will change by a small amount $\Delta\rho$ from equilibrium. This is accompanied by a pressure difference

$$\Delta P = P_1 - P_2,$$

where, to first order,

$$P_1 = P_0 + \frac{dP}{d\rho} \Delta\rho_1$$

and

$$P_2 = P_0 + \frac{dP}{d\rho} \Delta\rho_2.$$

Using Eq. (30.25) we have

$$\Delta P = \frac{dP}{d\rho} (\Delta\rho_1 - \Delta\rho_2) = v_s^2 (\Delta\rho_1 - \Delta\rho_2).$$

For a displacement x of the piston, and noting that the mass of the gas on either side of the piston does not change,

$$\Delta\rho_1 = \rho_1 - \rho_0 = \rho_0 \left(\frac{\rho_1}{\rho_0} - 1 \right) = \rho_0 \left(\frac{L}{L+x} - 1 \right)$$

and similarly

$$\Delta\rho_2 = \rho_2 - \rho_0 = \rho_0 \left(\frac{\rho_2}{\rho_0} - 1 \right) = \rho_0 \left(\frac{L}{L-x} - 1 \right).$$

Thus

$$\Delta P = v_s^2 \rho_0 \left(\frac{1}{1+x/L} - \frac{1}{1-x/L} \right)$$

so, to first order in x ,

$$\Delta P = -2v_s^2 \rho_0 \left(\frac{x}{L} \right).$$

Newton's second law for the piston of mass $m = 2LA\rho_0$ is then

$$m \frac{d^2 x}{dt^2} = A \Delta P$$

or

$$2LA\rho_0 \frac{d^2 x}{dt^2} = -2v_s^2 A\rho_0 \left(\frac{x}{L} \right).$$

Solving for the acceleration of the piston, we find the equation of motion to be

$$\frac{d^2x}{dt^2} = -\frac{v_s^2}{L^2}x.$$

The resulting simple harmonic motion of the piston,

$$x = x_0 \sin(\omega t),$$

has angular frequency

$$\omega = \frac{v_s}{L}. \quad (30.33)$$

This is in reasonable agreement with the fundamental frequency for an organ pipe closed at both ends, $\omega = \pi v_s/2L$. This shows that more extensive density fluctuations (larger L) will oscillate with a lower frequency.

Now we will add a uniform gravitational field of strength g directed in the positive- x direction. This represents the effect of a concentration of dark matter on the baryons in the photon–baryon fluid. Because dark matter is not subject to the intense radiation pressure of the CMB, it can form clumps capable of gravitationally assisting or resisting the motion of the fluid. Returning to our cylinder, Newton’s second law now gives

$$m \frac{d^2x}{dt^2} = A \Delta P + mg,$$

which leads to

$$\frac{d^2x}{dt^2} = -\frac{v_s^2}{L^2}x + g.$$

To solve this differential equation, define

$$y \equiv x - \frac{L^2}{v_s^2}g$$

so Newton’s second law becomes

$$\frac{d^2y}{dt^2} = -\frac{v_s^2}{L^2}y.$$

As before, the piston executes simple harmonic motion with angular frequency $\omega = v_s/L$, but now the oscillations are about $y = 0$, which corresponds to an equilibrium position of $x_{\text{eq}} = L^2 g/v_s^2 > 0$. This means that the compressions in Region 2, in the direction of the gravitational field, are of greater magnitude than are the rarefactions in Region 2. The equilibrium density in Region 2 (when $y = 0$) is

$$\rho_{2,\text{eq}} = \rho_0 \left(\frac{L}{L - x_{\text{eq}}} \right), \quad (30.34)$$

which is greater than ρ_0 , the average density of the gas in the cylinder.

We will take Region 2 to be a model of the oscillation of the photon–baryon fluid surrounding a local concentration of dark matter. The model indicates that the starting state of a density fluctuation is more likely to be one that would fall inward toward the dark matter rather than expand outward. The reason for this bias is that the quantum density fluctuations varied randomly between being underdense ($\rho_2 < \rho_0$ in our model) and overdense ($\rho_2 > \rho_0$) relative to the average density of the universe (ρ_0 in our model), and not relative to the density of the equilibrium configuration in the presence of dark matter ($\rho_{2,\text{eq}}$). Similarly, if the initial position of our model’s piston were chosen randomly about $x = 0$, we would be more likely to get a model for which Region 2 would initially move in the $+x$ direction (in the direction of the gravitational field) because $x_{\text{eq}} > 0$ introduces a bias. Therefore, in the early universe, collapsing initial configurations were favored over expanding ones. Because the compressions in the direction of the gravitational field are of greater magnitude than are the rarefactions, we expect the peaks of the angular power spectrum to be enhanced for odd harmonics (compressions), and diminished for even harmonics (rarefactions).

The Implications of the Angular Power Spectrum Peaks

As shown in Fig. 30.16, the first peak in the angular power spectrum of the CMB (Fig. 30.13) is due to the compression of a large region (large L) that reached its maximum compression at the time of decoupling. The first trough is produced by a smaller region (smaller L) that started oscillating earlier, when it became sub-horizon-sized. This smaller region was able to oscillate more quickly (larger ω , from Eq. 30.33), so that it arrived at $\delta T = 0$ at the

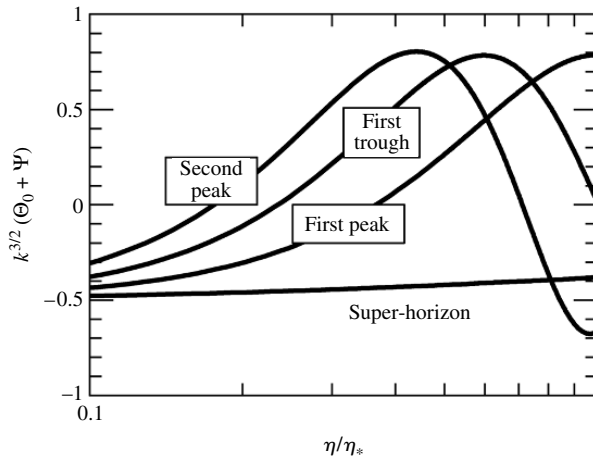


FIGURE 30.16 Several modes of the acoustic oscillations in the early universe. The sum of the potentials plotted on the vertical axis can be interpreted as δT , so positive values indicate a compression of the photon–baryon fluid and negative values indicate a rarefaction. The time η is plotted on the horizontal axis as a fraction of the time at recombination (η_*), so $\eta/\eta_* = 1$ at recombination. The oscillations are not symmetric about $\delta T = 0$ because the baryons are attracted to concentrations of dark matter. [Figure adapted from Dodelson, *Neutrinos, Flavor Physics, and Precision Cosmology: Fourth Tropical Workshop on Particle Physics and Cosmology*, José F. Nieves and Raymond R. Volkas (eds.), AIP Conference Proceedings, 689, 184, 2003.]

time of decoupling, where δT is the variation in temperature from equilibrium. The second peak is due to the oscillation of a still smaller region that passed through its maximum compression and then reached its maximum rarefaction at the time of decoupling. Note that the magnitude of δT at decoupling for the first peak is greater than the magnitude of δT at decoupling for the second peak. This is due to the biasing effect of the gravitational pull on the baryons by local concentrations of dark matter. As a result, δT for the compression is greater (in magnitude) than that for the rarefaction. Thus the height of the second peak is less than that of the first peak, and the relative suppression of the second peak increases as the value of Ω_b increases. Comparison of the heights of the first and second harmonic peaks reveals the density of baryonic matter in the universe.

The third peak (the third harmonic) is due to an oscillation that reaches its second maximum compression at the time of decoupling. Just as the relative heights of the first two peaks indicate the amount of baryonic matter in the universe, the height of the third peak is sensitive to the density of dark matter. In essence, the location and heights of the three peaks can be used to determine, with some accuracy, the values of the three density parameters Ω_0 , $\Omega_{b,0}$, and $\Omega_{\text{dark},0}$. The comparable heights of the second and third peaks indicate that most of the matter in the universe is in the form of dark matter.

The line labeled “Super-Horizon” in Fig. 30.16 describes the behavior of a density fluctuation so vast that it remained outside the particle horizon until recombination, so it never experienced acoustic oscillations. The CMB photons from such a super-horizon region were affected by both the primordial temperature fluctuation and the gravitational redshift of the Sachs-Wolfe effect as they climbed out of the gravitational potential wells. The angular size of these regions corresponds roughly to $\ell < 50$, so the left side of the angular power spectrum shows only a low, flat plain for these values of ℓ .

Figure 30.17 shows that on the right side of the angular power spectrum, the peaks die off for $\ell > 1000$. This is the Silk damping that occurs when the random-walking photons can diffuse between the compressions and rarefactions of the short-wavelength sound waves and so decrease their amplitudes.

As you admire the harmonic structure of these celestial sound waves, it is striking to realize that the very existence of harmonics strongly supports the idea of inflation. The regularity in this harmonic structure means that the oscillations of all density fluctuations of a given size (and so having the same frequency) must reach their maximum compressions and rarefactions *at the same time*. This requires that they begin their oscillations simultaneously. Inflation provides the natural mechanism of generating density fluctuations and expanding them to super-horizon size. They remain “frozen” until the particle horizon grows large enough to encompass them and allow them to oscillate. Without this inflationary mechanism, a collection of density fluctuations that was generated randomly in time would not be in phase, and a series of evenly spaced peaks would not be observed in the power spectrum.

Figure 30.18 shows the effect on the power spectrum of varying several cosmological parameters. You may find it illuminating to download the CMBFAST program (listed in the references) that will quickly and accurately calculate the angular power spectrum for a choice of model input parameters. Future missions are being planned to better resolve the higher harmonic peaks and so further constrain the models. Some theoretical models of structure formation have already been decisively eliminated by comparing them with the observed angular power spectrum. For example, exotic relics of the false vacuum such as

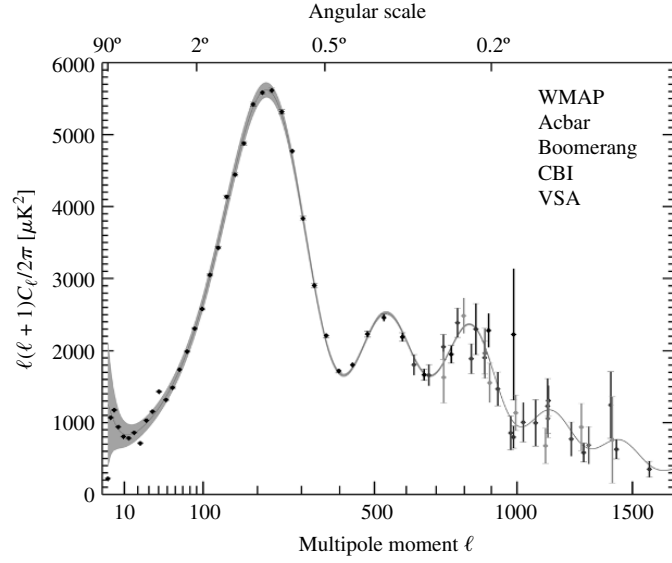


FIGURE 30.17 The results of recent measurements of the angular power system by a variety of projects. The solid line is the best fit to the WMAP data. [Figure adapted from Hinshaw et al., *Ap. J.* (submitted), 2006, and courtesy of NASA and the WMAP Science Team.]

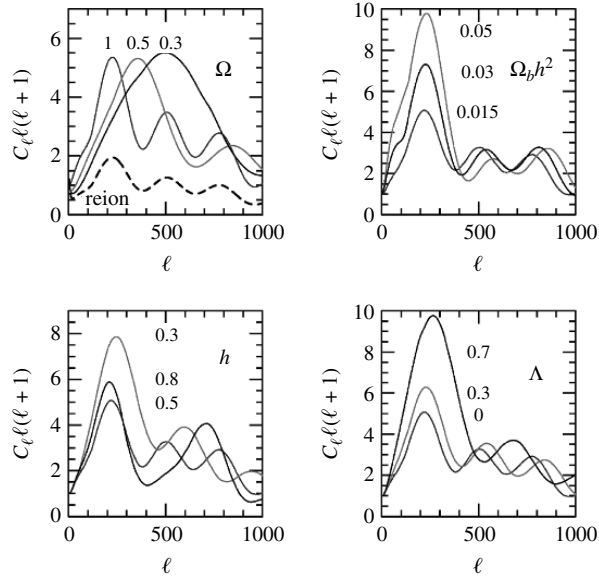


FIGURE 30.18 Theoretical calculations of the angular power spectrum for a variety of cosmological parameters. For the lower right panel, $\Omega_0 = 1$. (Figure adapted from Kamionkowski and Kosowsky, *Annu. Rev. Nucl. Part. Sci.*, 49, 77, 1999.)

monopoles and domain walls were once considered candidates for the “seeds” of structure formation. However, their power spectrum is quite different from the observed spectrum. These relics may or may not exist, but they do not play a significant role in the growth of structure.

Polarization of the CMB Anisotropies

In March 2006, the latest WMAP findings were announced, including an analysis of the polarization of the CMB anisotropies. The magnitude of the polarization signal is less than 1% of the size of the temperature fluctuations, so this was an extremely demanding measurement. Complications such as polarized foreground emission due to Galactic synchrotron and thermal dust emission had to be carefully accounted for. The polarization originates in the Thompson scattering of the CMB photons by the free electrons in the photon–baryon fluid. The top of Fig. 30.19 shows how the electric field of incident unpolarized radiation (in the $+y$ direction) causes an electron to vibrate. This produces scattered light in the $+x$ and $+z$ directions that is linearly polarized. However, for an isotropic radiation field coming at the electron from all directions, there will be no net polarization. The observed partial polarization of the CMB therefore requires a radiation field that is more intense along one axis than along another axis. This is shown at the bottom of Fig. 30.19, where the incident

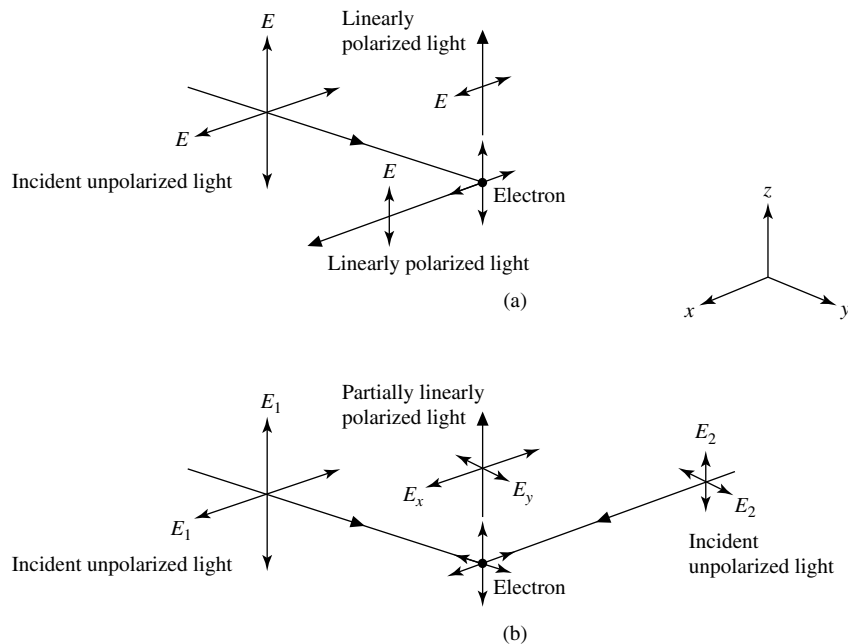


FIGURE 30.19 Polarization of light by Thompson scattering from an electron. In (a), incident unpolarized light from one direction produces linearly polarized light upon scattering, when observed in the x – z plane. In (b), incident unpolarized light of different intensities ($E_1 > E_2$) from two directions produces partially linearly polarized light ($E_x > E_y$) when observed along the z -axis.

light in the $+y$ direction is more intense than that in the $+x$ direction ($E_1 > E_2$). We say that such a radiation field has a *quadrupole moment*. As a result, the scattered radiation along the z -axis is partially polarized in the x direction.

If Thompson scattering occurs frequently, then the photon directions quickly become randomized, and the quadrupole moment of the radiation field cannot be sustained. This means that the observed polarization must be established at the time of decoupling, when the universe became transparent and Thompson scattering ceased. After decoupling there are no free electrons to scatter the CMB photons until the first generation of stars reionizes the universe. This occurs at a much lower redshift than decoupling ($[z_{\text{dec}}]_{\text{WMAP}} = 1089$), so the signature of reionization can be gleaned from the polarization measurements. The results show that reionization took place at a redshift of $z = 10.9^{+2.7}_{-2.3}$.

There are two ways of producing a local quadrupole moment in the CMB: by the flow of the photon–baryon fluid and by the effect of gravitational radiation. The acoustic oscillation of the photon–baryon fluid implies that there is a velocity gradient within the fluid. For simplicity, consider the situation where the fluid is moving radially inward toward a local concentration of dark matter (see Fig. 30.20). From the frame of reference of an electron in the fluid, the neighboring elements of fluid are all moving toward it, with (in this example) the fluid in the radial direction approaching more rapidly than the fluid in the transverse direction. The relativistic headlight effect (Section 4.3) then causes the radiation from the radial directions to be more intense than the radiation in the transverse directions. This produces the quadrupole moment in the local radiation field that is responsible for partially polarizing the photons as they are scattered by the electron. The polarization pattern (the direction of the observed electric field of the CMB photons) is directed radially from a cooler temperature fluctuation ($\Delta T < 0$) and forms tangential loops around a warmer fluctuation ($\Delta T > 0$). This pattern is called an *E-mode*. [Notice in Fig. 30.19(b) that the electric field of the partially polarized light is along the direction of the less intense source of light and perpendicular to the direction of the more intense source.]

The second means of creating a local quadrupole moment in the CMB is through the effect of the gravitational radiation produced by the primordial density fluctuations. When a gravitational wave passes through the photon–baryon fluid, the space containing the

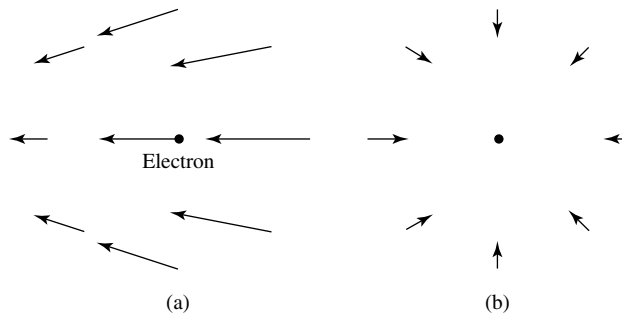


FIGURE 30.20 (a) The motion of an electron in the photon–baryon fluid as the fluid moves radially inward toward its maximum compression. (b) The velocity field seen from the rest frame of the electron.

photons will be stretched along one axis while it is compressed along a perpendicular axis. The resulting crowding and thinning of the photons produce a quadrupole moment in the radiation field, and hence the observed partial polarization of the CMB. In addition to the E-mode pattern of polarization, gravitational waves can produce a curling pattern called *B-modes* that spirals outward from the temperature fluctuations (the direction of the spiral depending on the sign of the temperature fluctuation). Any polarization pattern can be decomposed into its E-mode and B-mode contributions, so observations of the polarization of the CMB carry much information about the conditions at the time of decoupling.

Figure 30.21 shows the WMAP measurements of the CMB polarization in five frequency bands. First of all, note that the vertical axis is logarithmic and displays the *square root* of the power spectrum displayed in Fig. 30.13. For $\ell > 50$ (beyond the low, flat plain of the Sachs–Wolfe effect) we expect to see a correlation between the TT and EE curves. This is because the velocity gradients of the acoustic oscillations should be greatest for modes that are near their equilibrium configurations, rather than at their maximum compression or rarefaction, at the time of decoupling. (For a harmonic wave, the largest velocity gradients are found where the wave displacement is a maximum, while the maximum compression or rarefaction of the medium occurs at the points of zero displacement.) Thus the peaks in the angular power spectrum should be out of phase with the E-mode peaks. For example, the

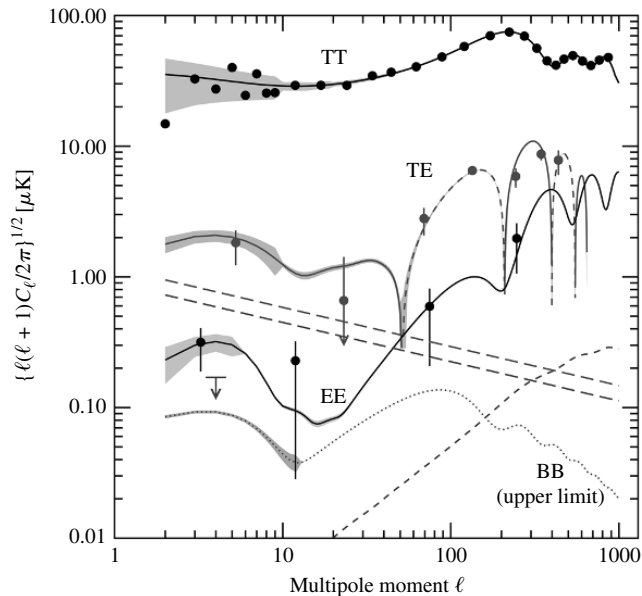


FIGURE 30.21 Plots of the WMAP measurements of the angular power spectrum (TT), the temperature/E-mode correlation (TE), the E-mode polarization (EE), and the B-mode polarization (BB). WMAP found no evidence for B-modes, so the upper limit for them is shown as a dotted line. The curves are for the best-fit cosmological model. [Figure adapted from Page et al., *Ap. J.* (submitted), 2006, and courtesy of NASA and the WMAP Science Team.]

first peak in the angular power spectrum should correspond to a minimum in the E-mode curve, and this is indeed what is observed. The curve labeled “TE” shows the expected correlation between the temperature and E-mode observations.

To date, WMAP has found no evidence for the B-modes, but their expected signal is faint and the WMAP observatory and the European Space Agency’s Planck spacecraft (planned to launch in 2007) will continue to search for the B-modes. A definitive measurement of the B-modes for $\ell < 100$ would provide insights about the physics of the early universe at energies of 10^{15} – 10^{16} GeV, back to the end of inflation.

The detection of B-modes would also impact an alternative cosmology, the cyclic model. This radical (but testable) alternative to the Big Bang/inflationary scenario proposes a cyclic universe in which the large-scale structure of the universe was set up *before* the Big Bang, during a slow contraction phase that followed an even earlier expansion. The observed energy we attribute to the Big Bang was instead released by the collision of two universes residing on two parallel three-dimensional membranes separated by a fourth dimension. The Big Bang/inflation model and the cyclic model offer the only possible mechanisms for producing nearly scale-invariant, adiabatic fluctuations (as are observed by WMAP). The collision of membranes is incapable of generating B-modes in the polarization of the CMB, so the detection of B-modes would rule out this imaginative model and at the same time confirm the inflationary model.¹⁶

Evidence of Harmonics in Large-Scale Structure

If the quantum density fluctuations, amplified by concentrations of dark matter, really are the “seeds” of structure formation in the universe, and if the density fluctuations reveal an acoustic harmonic structure, then it seems reasonable to ask if that harmonic structure was imprinted on the large-scale structure of the universe. Incredibly, the answer is yes. A large sample of 46,748 galaxies from the Sloan Digital Sky Survey was examined to see if there was any correlation in their positions. There was a slight excess of galaxies separated by $100h^{-1}$ Mpc, or about 140 Mpc for $[h]_{\text{WMAP}} = 0.71$. This is in excellent agreement with the separation predicted for the first harmonic sound wave when the growth of structure due to the universal expansion is taken into account. Astronomers have another powerful statistical tool to apply to the secrets of the early universe.

Because the location of the first peak is sensitive to the value of $\Omega_0 = \Omega_{m,0} + \Omega_{\Lambda,0}$, we can plot the WMAP results on the $\Omega_{m,0}$ – $\Omega_{\Lambda,0}$ plane together with the results from high- z supernovae (recall Fig. 29.28). We also add the results of studies of X-rays from clusters of galaxies by the Chandra orbiting observatory, which, when combined with gravitational lens data, allow $\Omega_{m,0}$ to be determined. The result, shown in Fig. 30.22, displays a beautiful concordance—a spectacular agreement with the WMAP values obtained by using theoretical models to reproduce the observed angular power spectrum.

The science of precision cosmology has arrived.

¹⁶The cyclic model does not deny the expansion of the universe from a hot, dense state 13.7 billion years ago. If evidence favoring the cyclic model is discovered, only Chapter 30 of this book will have to be rewritten. You are referred to Steinhardt and Turok (2002) for details of their cyclic model of the universe.

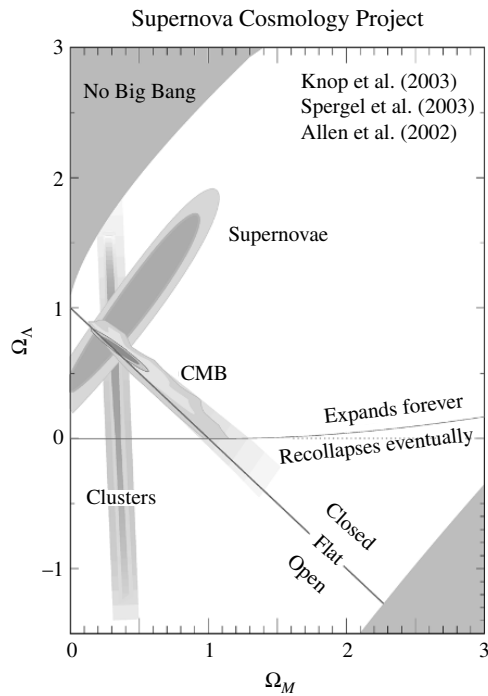


FIGURE 30.22 The cosmic concordance of the results of three different observing projects on the $\Omega_{m,0}$ – $\Omega_{\Lambda,0}$ plane. (Adapted from a figure courtesy of the Supernova Cosmology Project.)

The Question

In our astrophysics class, a student once asked, “Why are we here?” The answer is as amazing to us as it was to the class:

We are here because, 13.7 billion years ago, the universe borrowed energy from the vacuum to create vast amounts of matter and antimatter in nearly equal numbers. Most of it annihilated and filled the universe with photons. Less than one part per billion survived to form protons and neutrons and then the hydrogen and helium that make up most of everything there is. Some of this hydrogen and helium collapsed to make the first generation of massive stars, which produced the first batch of heavy elements in their central nuclear fires. These stars exploded and enriched the interstellar clouds that would form the next generation of stars. Finally, about 4.6 billion years ago, one particular cloud in one particular galaxy collapsed to form our Sun and its planetary system. Life arose on the third planet, based on the hydrogen, carbon, nitrogen, oxygen, and other elements found in the protostellar cloud. The development of life transformed Earth’s atmosphere and allowed life to move onto land. Sixty-five million years ago, a fortunate collision with a large meteoroid hastened the demise of the dinosaurs and allowed small, furry mammals to take center stage. Primitive men and women evolved and moved out of Africa to conquer the world with their new knowledge of tools, language, and agriculture. After raising food on the land, your ancestors, your parents, and then you consumed this food and breathed the air. Your own body is a collection of the atoms that were created billions of years earlier in the interiors of stars, the fraction of a fraction of a percent of normal matter that escaped annihilation in the first microsecond of the universe. Your life and everything in the world around you is intimately tied to countless aspects of modern astrophysics.

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PROBLEMS

30.1 In Problem 9.3, you derived Eq. (9.65) for the number density of blackbody photons:

$$n = \frac{u}{2.70kT} = \frac{aT^3}{2.70k}.$$

Use this result along with the baryon density, $\rho_{b,0}$, to estimate the ratio of the number of baryons to the number of photons in today's universe. For convenience, assume that the universe is made solely of hydrogen.

- 30.2** (a) Suppose observations of the microlensing of stars in the Large Magellanic Cloud showed that most of the dark matter in the Milky Way's dark halo is in the form of ordinary brown dwarfs and Jupiter-sized objects. Explain whether this would favor hot or cold dark matter for the nonbaryonic matter in the universe, and why.
- (b) In fact, observations indicate that this is probably *not* the case: Less than 20% of the Milky Way's dark halo is in the form of ordinary brown dwarfs and Jupiter-sized objects. How do these observations change the conclusion you reached in part (a)?
- 30.3** An example of spontaneous symmetry breaking can be obtained by considering a small ball of mass $m = 1/9.8$ kg that is free to roll on a surface whose height (in meters) is given by

$$h(x) = kx^2 + \varepsilon x^4,$$

where $k = \pm 1 \text{ m}^{-1}$ and $\varepsilon = 0.5 \text{ m}^{-3}$. The gravitational potential energy of the ball is then $V(x) = mgh(x)$.

- (a) Make two graphs of $V(x)$ from $x = -2 \text{ m}$ to $x = 2 \text{ m}$, one for each choice of sign for k .
- (b) The case of $k = 1 \text{ m}^{-1}$ corresponds to the symmetric false vacuum. Where is the point of equilibrium? Is this point stable or unstable? (In the case of stable equilibrium, if the ball is displaced slightly, it will return to the equilibrium point.)
- (c) The case of $k = -1 \text{ m}^{-1}$ corresponds to the broken symmetry of the true vacuum. Where are the three points of equilibrium? Which of these points are stable, and which are unstable?
- (d) For the case of $k = -1 \text{ m}^{-1}$, consider the ball at rest at the origin. What are the implications of the uncertainty principle, Eq. (5.20), for the ball remaining in that position? In what ways is this situation analogous to that of the supercooled false vacuum just before inflation?
- 30.4** Compare the energy density of CMB photons at the end of the GUTs epoch to that of the false vacuum.
- 30.5** According to the standard Big Bang cosmology, what was the value of the scale factor, R , in the radiation era at the Planck time? When inflation is considered (take $t/\tau_i = 100$), what was

the value of the scale factor at the Planck time? For both situations, what was the size of the presently observable universe at the Planck time?

- 30.6** Estimate the thickness of a typical cosmic string.
- 30.7** (a) Estimate the mass of baryonic matter contained within a causally connected region (diameter = the horizon distance) when $T = 10^9$ K during the radiation era. Express your answer in solar units. How does this compare with the Jeans mass shown in Fig. 30.7?
 (b) Show that during the radiation era, the mass of the baryonic matter inside this causally connected region varied as T^{-3} . What does this say about the relative magnitudes of the region's mass and the Jeans mass throughout the radiation era?
- 30.8** Estimate the mass of baryonic matter contained within a causally connected region at the time of the transition from the radiation era to the matter era. Express your answer in solar units. (Use WMAP values.) From the magnitude of your answer, did most of the baryonic density fluctuations become sub-horizon-sized during the radiation era or during the matter era?
- 30.9** Show that if a baryonic density perturbation at the present time is $(\delta\rho/\rho)_0 = 1$, then its value at a redshift z in the matter era was $\delta\rho/\rho = (1+z)^{-1}$. By what factor does any $\delta\rho/\rho < 1$ increase between a redshift z_1 and a redshift z_2 ($z_1 > z_2$) in the matter era? You may assume that the universe is nearly flat and that $z < 1100$ for baryonic fluctuations.
- 30.10** Follow the linearization procedure of Example 14.3.1 and show that the variations in the temperature of the CMB are related to the baryonic density fluctuations by Eq. (30.30).
- 30.11** Assuming a flat universe, find the angular size of the largest causally connected region of the CMB. *Hint:* We see this region as it existed at the time of decoupling, when the CMB photons were set free.
- 30.12** Estimate the value of ℓ when Silk damping becomes important for the angular power spectrum of the CMB anisotropies.
- 30.13** According to the WMAP CMB polarization results, how old was the universe when the first stars ignited?

APPENDIX

C

Solar System Data

Planetary Physical Data						
Planet	Mass ^a ($M_{\oplus}$)	Equatorial Radius ^b ($R_{\oplus}$)	Average Density (kg m^{-3})	Sidereal Rotation Period (d)	Oblateness ($R_e - R_p$)/ R_e	Bond Albedo
Mercury	0.05528	0.3825	5427	58.6462	0.00000	0.119
Venus	0.81500	0.9488	5243	243.018	0.00000	0.750
Earth	1.00000	1.0000	5515	0.997271	0.0033396	0.306
Mars	0.10745	0.5326	3933	1.02596	0.006476	0.250
Jupiter	317.83	11.209	1326	0.4135	0.064874	0.343
Saturn	95.159	9.4492	687	0.4438	0.097962	0.342
Uranus	14.536	4.0073	1270	0.7183	0.022927	0.300
Neptune	17.147	3.8826	1638	0.6713	0.017081	0.290
Pluto	0.0021	0.178	2110	6.3872	0.0000	0.4 – 0.6
2003 UB313	0.002?	0.188	2100?			0.6?

Planetary Orbital and Satellite Data						
Planet	Semimajor Axis (AU)	Orbital Eccentricity	Sidereal Orbital Period (yr)	Orbital Inclination to Ecliptic (°)	Equatorial Inclination to Orbit (°)	Number Natural Satellites
Mercury	0.3871	0.2056	0.2408	7.00	0.01	0
Venus	0.7233	0.0067	0.6152	3.39	177.36	0
Earth	1.0000	0.0167	1.0000	0.000	23.45	1
Mars	1.5236	0.0935	1.8808	1.850	25.19	2
Jupiter	5.2044	0.0489	11.8618	1.304	3.13	63
Saturn	9.5826	0.0565	29.4567	2.485	26.73	47
Uranus	19.2012	0.0457	84.0107	0.772	97.77	27
Neptune	30.0476	0.0113	164.79	1.769	28.32	13
Pluto	39.4817	0.2488	247.68	17.16	122.53	3
2003 UB313	67.89	0.4378	559	43.99		1

^a $M_{\oplus} = 5.9736 \times 10^{24} \text{ kg}$

^b $R_{\oplus} = 6.378136 \times 10^6 \text{ m}$

Data of Selected Major Satellites						
Satellite	Parent Planet	Mass (10^{22} kg)	Radius (10^3 km)	Density (kg m^{-3})	Orbital Period (d)	Semimajor Axis (10^3 km)
Moon	Earth	7.349	1.7371	3350	27.322	384.4
Io	Jupiter	8.932	1.8216	3530	1.769	421.6
Europa	Jupiter	4.800	1.5608	3010	3.551	670.9
Ganymede	Jupiter	14.819	2.6312	1940	7.155	1070.4
Callisto	Jupiter	10.759	2.4103	1830	16.689	1882.7
Titan	Saturn	13.455	2.575	1881	15.945	1221.8
Triton	Neptune	2.14	1.3534	2050	5.877	354.8